

```
In [1]: import numpy as np
import pandas as pd
```

```
In [2]: import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

```
In [3]: import seaborn as sns
import matplotlib.pyplot as plt
import scipy.stats as st
%matplotlib inline

sns.set(style="whitegrid")
```

```
In [4]: import warnings
warnings.filterwarnings('ignore')
```

```
In [5]: df = pd.read_csv(r'C:\Users\DELL\Downloads\24th, 25th- Advanced EDA project\EDA-
```

```
In [6]: df
```

```
Out[6]:
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	tl
0	63	1	3	145	233	1	0	150	0	2.3	0	0	
1	37	1	2	130	250	0	1	187	0	3.5	0	0	
2	41	0	1	130	204	0	0	172	0	1.4	2	0	
3	56	1	1	120	236	0	1	178	0	0.8	2	0	
4	57	0	0	120	354	0	1	163	1	0.6	2	0	
...	...	...	...	...	...	...	...	...	...	...	...	...	...
298	57	0	0	140	241	0	1	123	1	0.2	1	0	
299	45	1	3	110	264	0	1	132	0	1.2	1	0	
300	68	1	0	144	193	1	1	141	0	3.4	1	2	
301	57	1	0	130	131	0	1	115	1	1.2	1	1	
302	57	0	1	130	236	0	0	174	0	0.0	1	1	

303 rows × 14 columns



```
In [7]: # print the shape
print('The shape of the dataset : ', df.shape)
```

The shape of the dataset : (303, 14)

```
In [8]: df.head()
```

Out[8]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2



In [9]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age         303 non-null    int64
1   sex         303 non-null    int64
2   cp          303 non-null    int64
3   trestbps    303 non-null    int64
4   chol        303 non-null    int64
5   fbs         303 non-null    int64
6   restecg     303 non-null    int64
7   thalach     303 non-null    int64
8   exang       303 non-null    int64
9   oldpeak     303 non-null    float64
10  slope       303 non-null    int64
11  ca          303 non-null    int64
12  thal        303 non-null    int64
13  target      303 non-null    int64
dtypes: float64(1), int64(13)
memory usage: 33.3 KB
```

In [10]: `df.dtypes`

```
Out[10]: age         int64
sex         int64
cp          int64
trestbps    int64
chol        int64
fbs         int64
restecg     int64
thalach     int64
exang       int64
oldpeak     float64
slope       int64
ca          int64
thal        int64
target      int64
dtype: object
```

In [11]: `df.describe()`

Out[11]:

	age	sex	cp	trestbps	chol	fbs	restecg
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.148515	0.528000
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.356198	0.525000
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.000000	1.000000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000

In [12]: `df.columns`

Out[12]: Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach', 'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'], dtype='object')

In [13]: `df['target'].nunique()`

Out[13]: 2

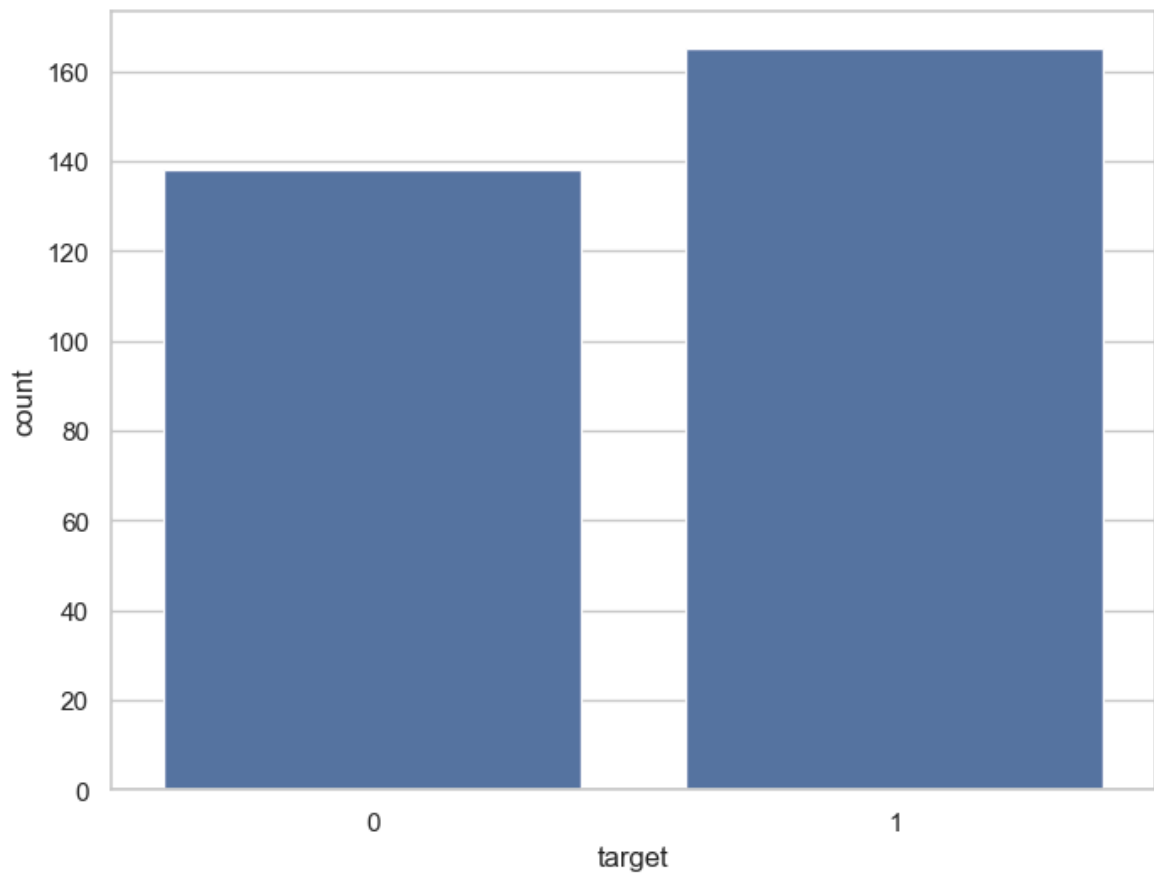
In [14]: `df['target'].unique()`

Out[14]: array([1, 0], dtype=int64)

In [15]: `df['target'].value_counts()`

Out[15]: target
 1 165
 0 138
 Name: count, dtype: int64

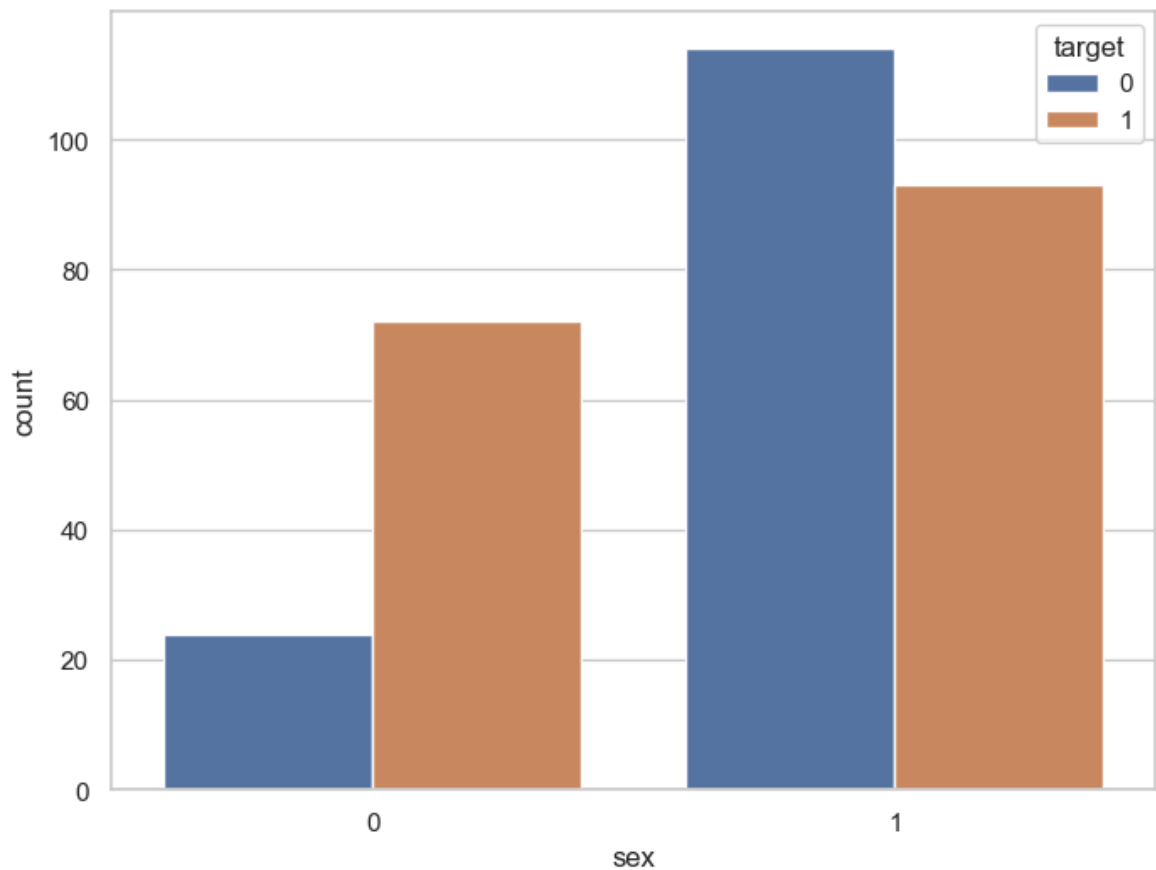
In [16]: `f, ax = plt.subplots(figsize=(8, 6))
 ax = sns.countplot(x="target", data=df)
 plt.show()`



```
In [17]: df.groupby('sex')['target'].value_counts()
```

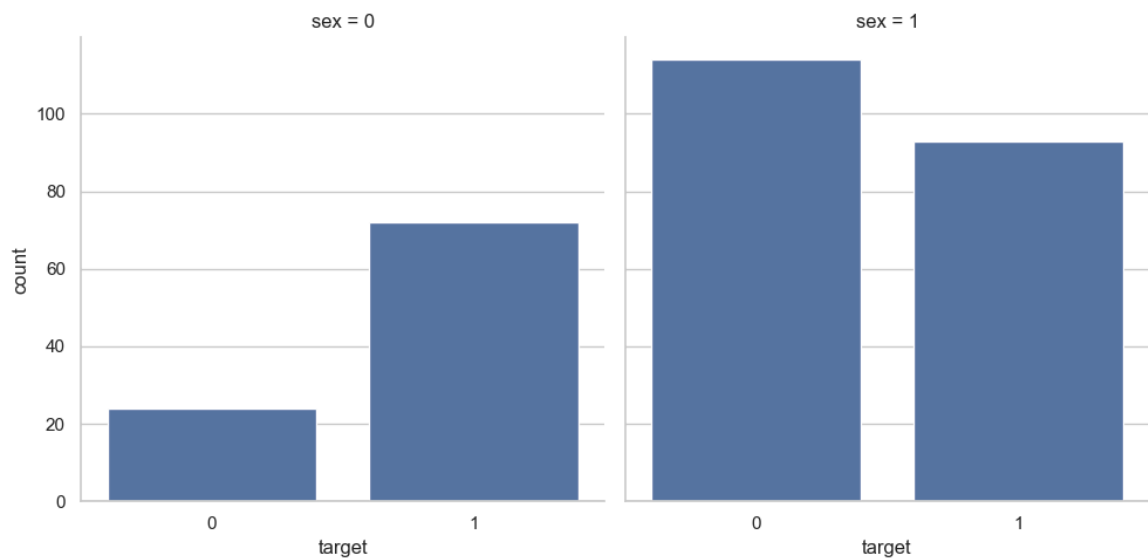
```
Out[17]: sex  target
0      1      72
      0      24
1      0     114
      1      93
Name: count, dtype: int64
```

```
In [18]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="sex", hue="target", data=df)
plt.show()
```

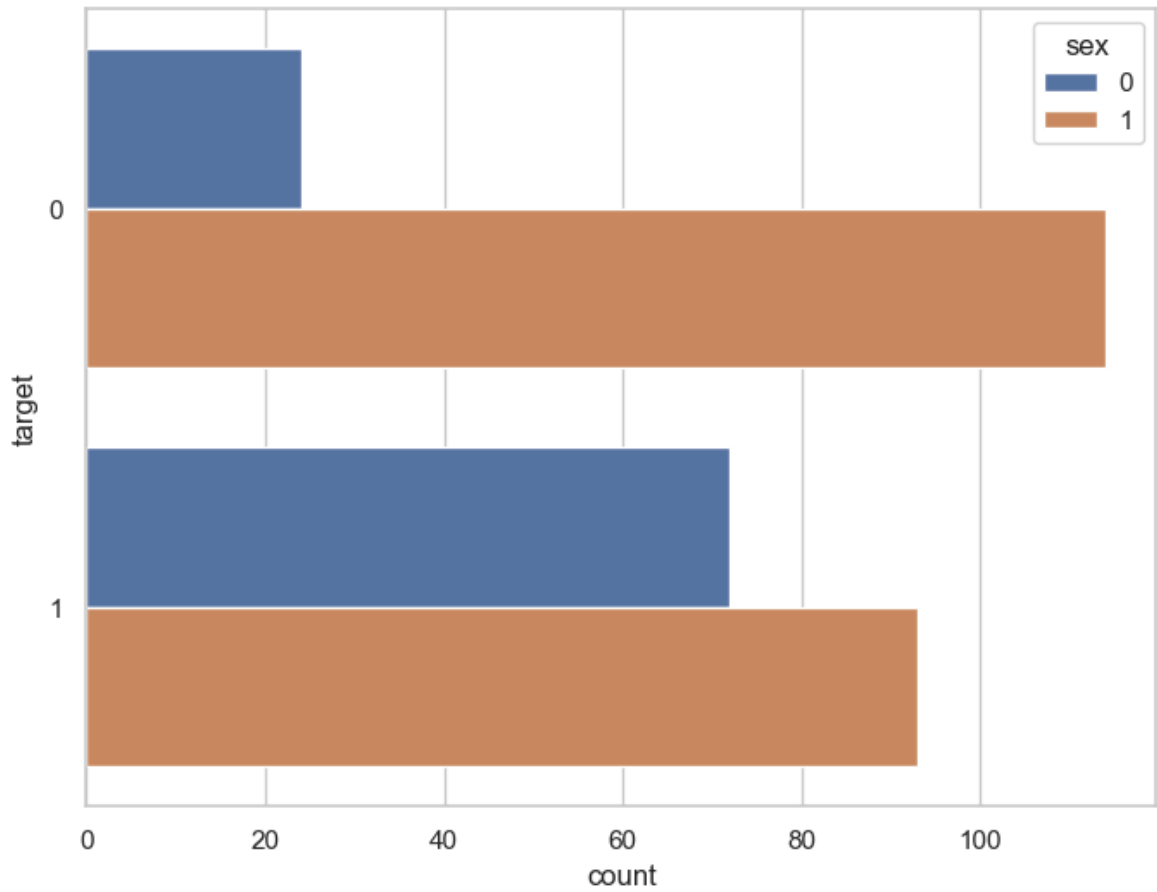


```
In [19]: ax = sns.catplot(x="target", col="sex", data=df, kind="count", height=5, aspect=
```

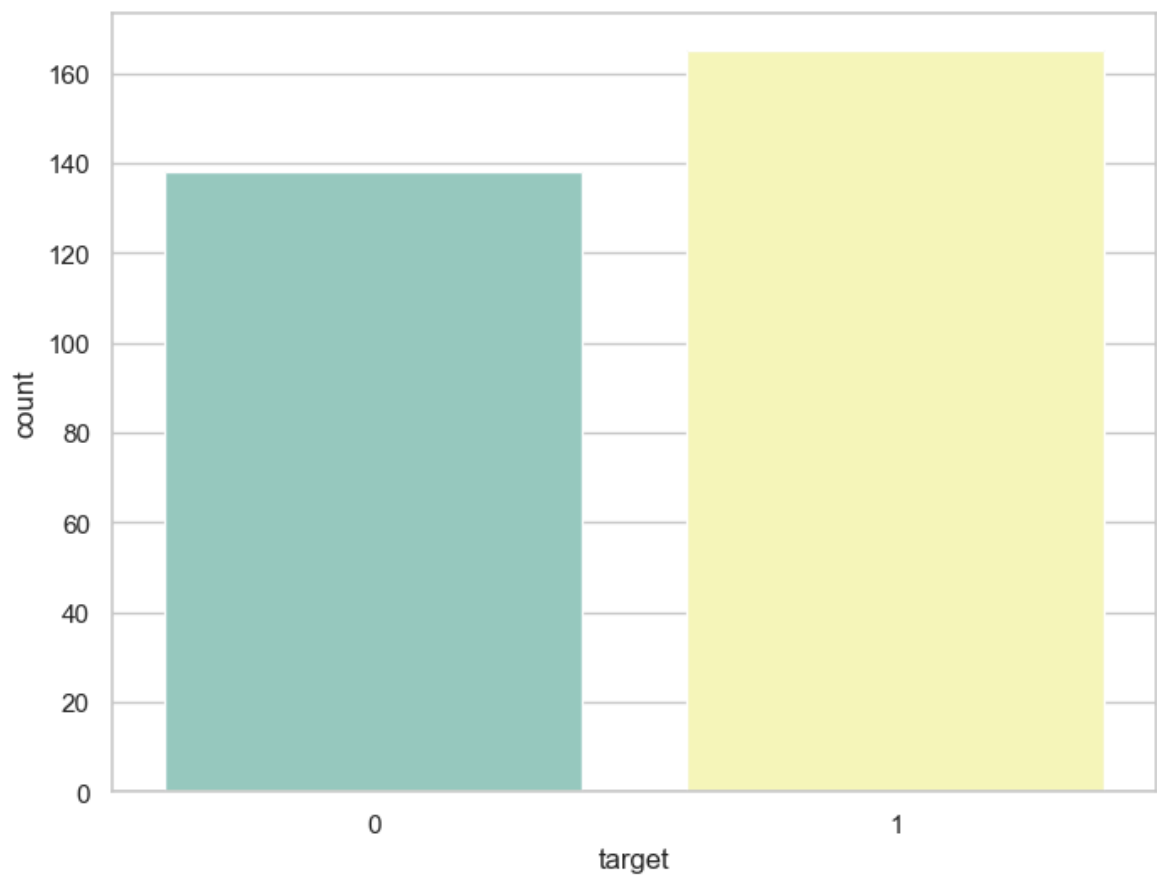
```
plt.show()
```



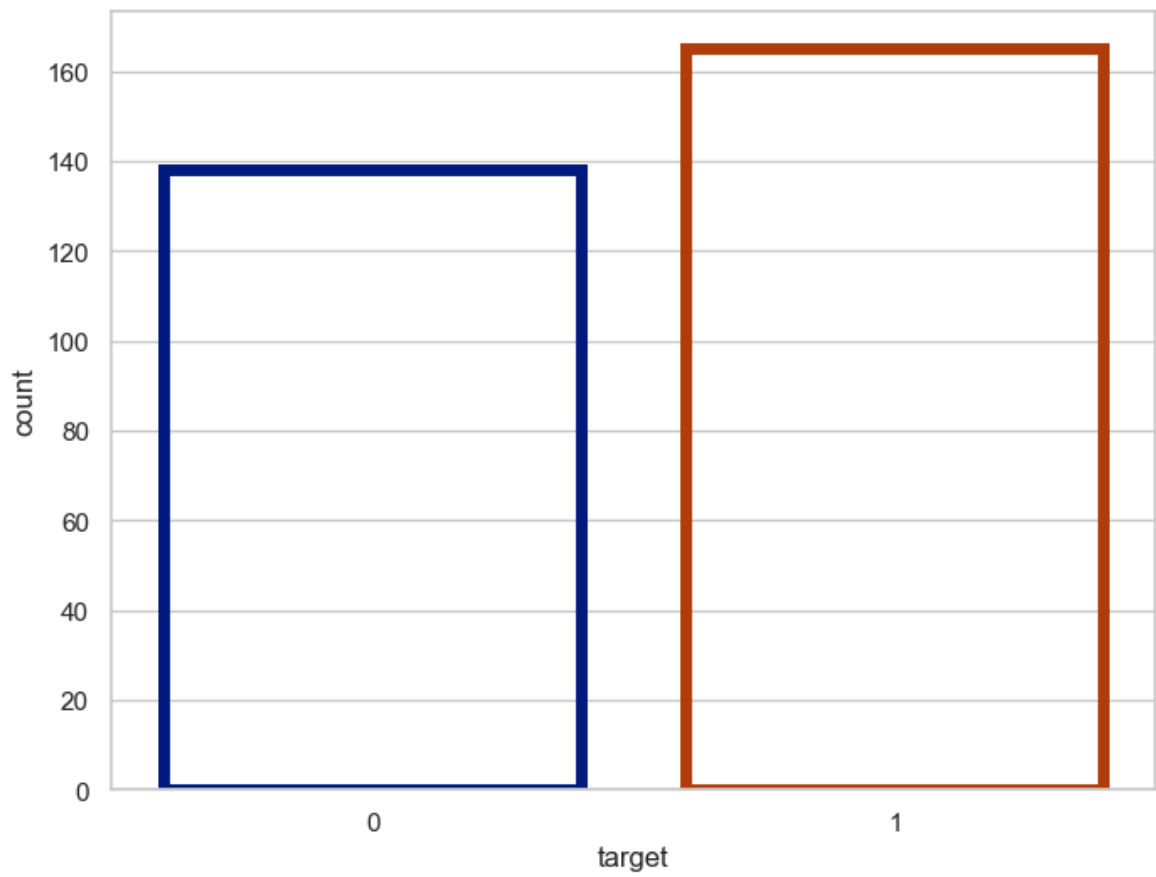
```
In [21]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.countplot(y="target", hue="sex", data=df)  
plt.show()
```



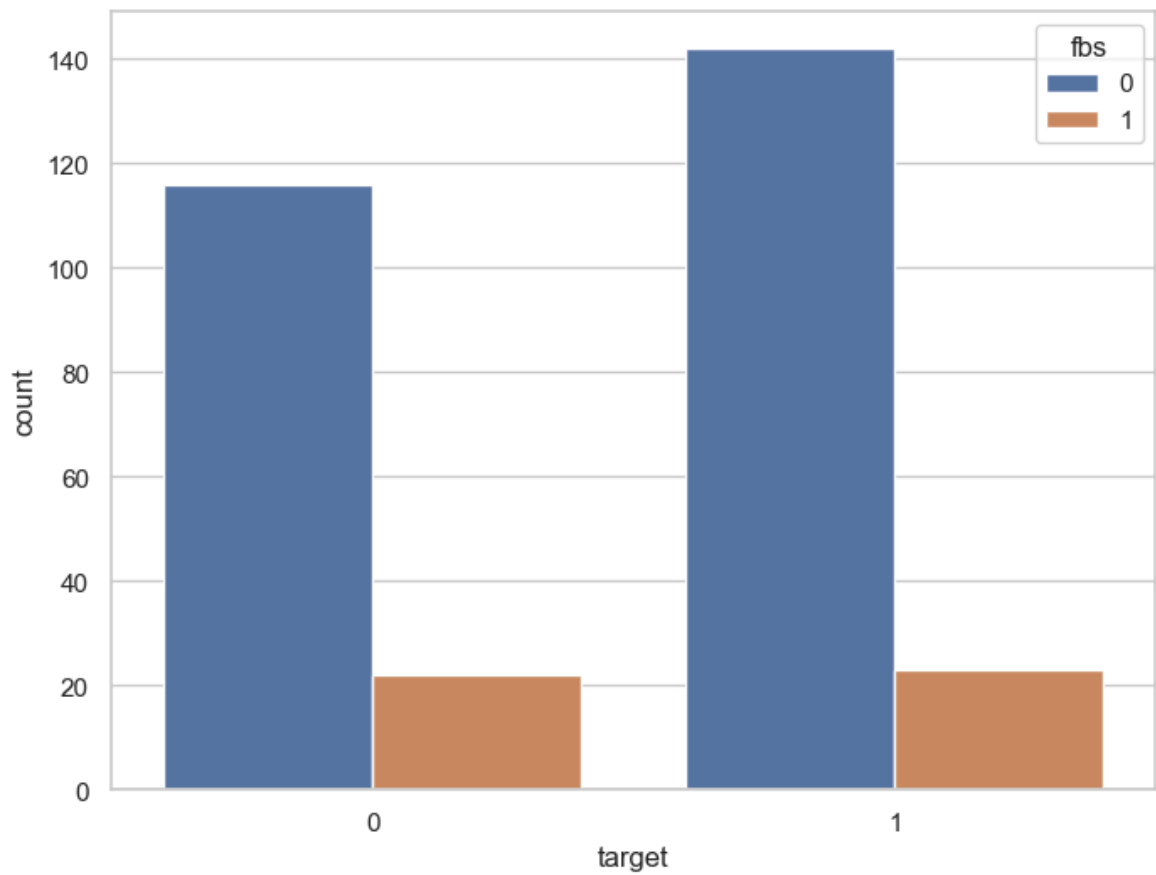
```
In [22]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.countplot(x="target", data=df, palette="Set3")  
plt.show()
```



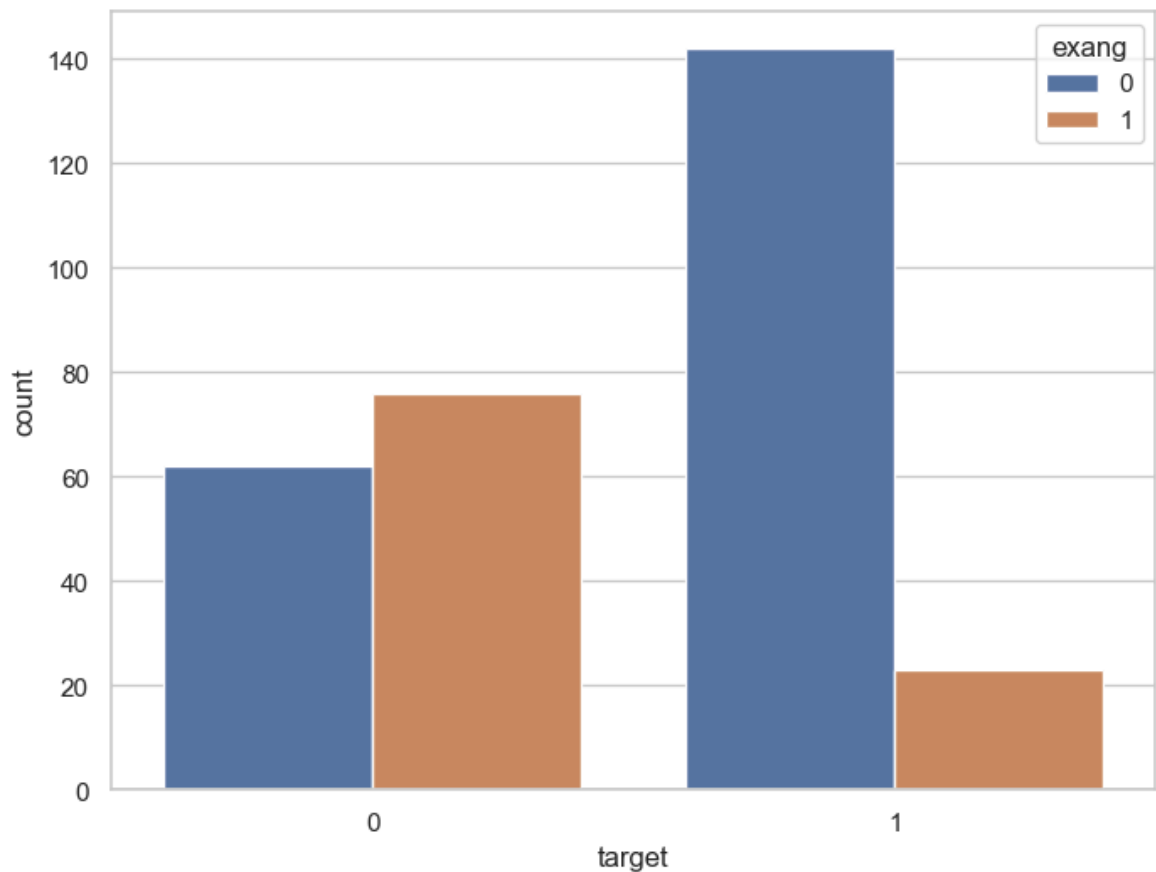
```
In [23]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="target", data=df, facecolor=(0, 0, 0, 0), linewidth=5, edge
plt.show()
```



```
In [24]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="target", hue="fbs", data=df)
plt.show()
```



```
In [25]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.countplot(x="target", hue="exang", data=df)  
plt.show()
```



```
In [26]: correlation = df.corr()
```



```
In [27]: correlation['target'].sort_values(ascending=False)
```

```
Out[27]: target      1.000000  
cp          0.433798  
thalach     0.421741  
slope       0.345877  
restecg     0.137230  
fbs         -0.028046  
chol        -0.085239  
trestbps    -0.144931  
age         -0.225439  
sex         -0.280937  
thal        -0.344029  
ca          -0.391724  
oldpeak     -0.430696  
exang       -0.436757  
Name: target, dtype: float64
```

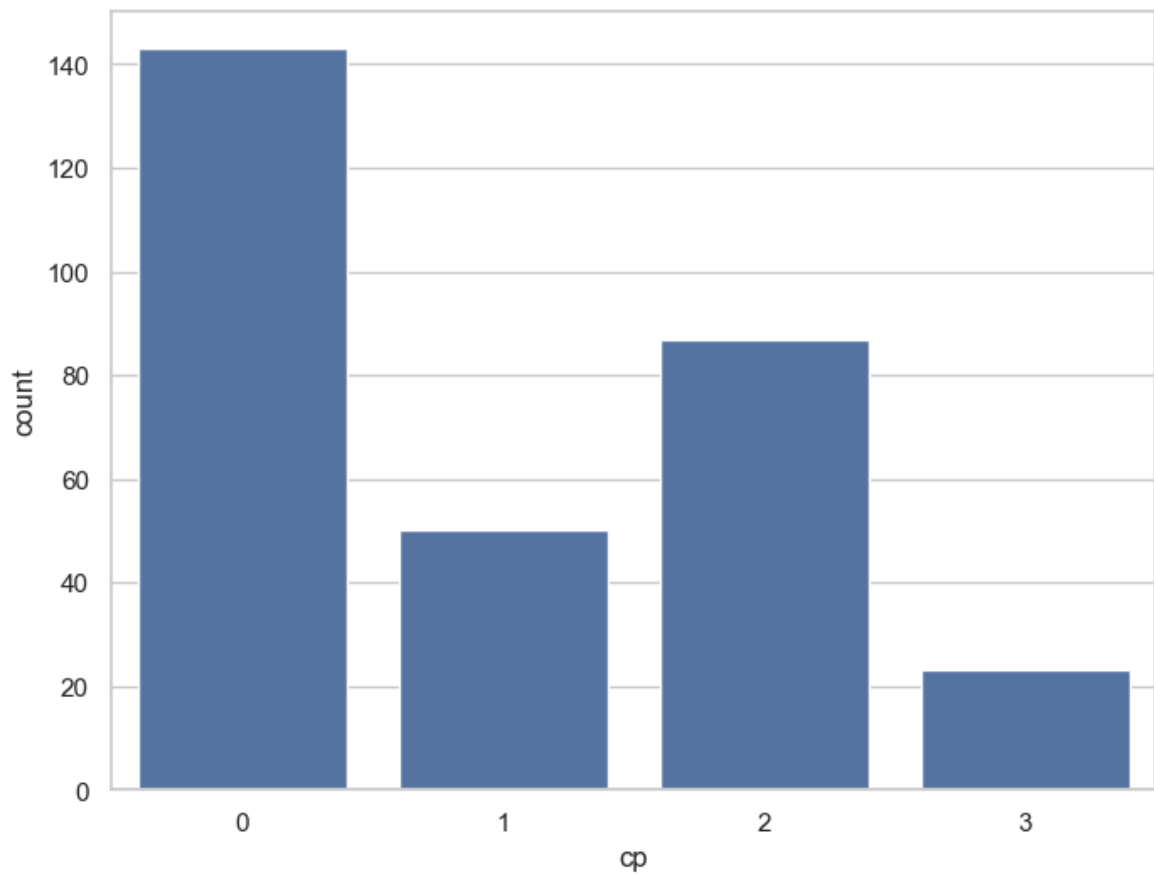
```
In [28]: df['cp'].nunique()
```

```
Out[28]: 4
```

```
In [29]: df['cp'].value_counts()
```

```
Out[29]: cp  
0      143  
2       87  
1       50  
3       23  
Name: count, dtype: int64
```

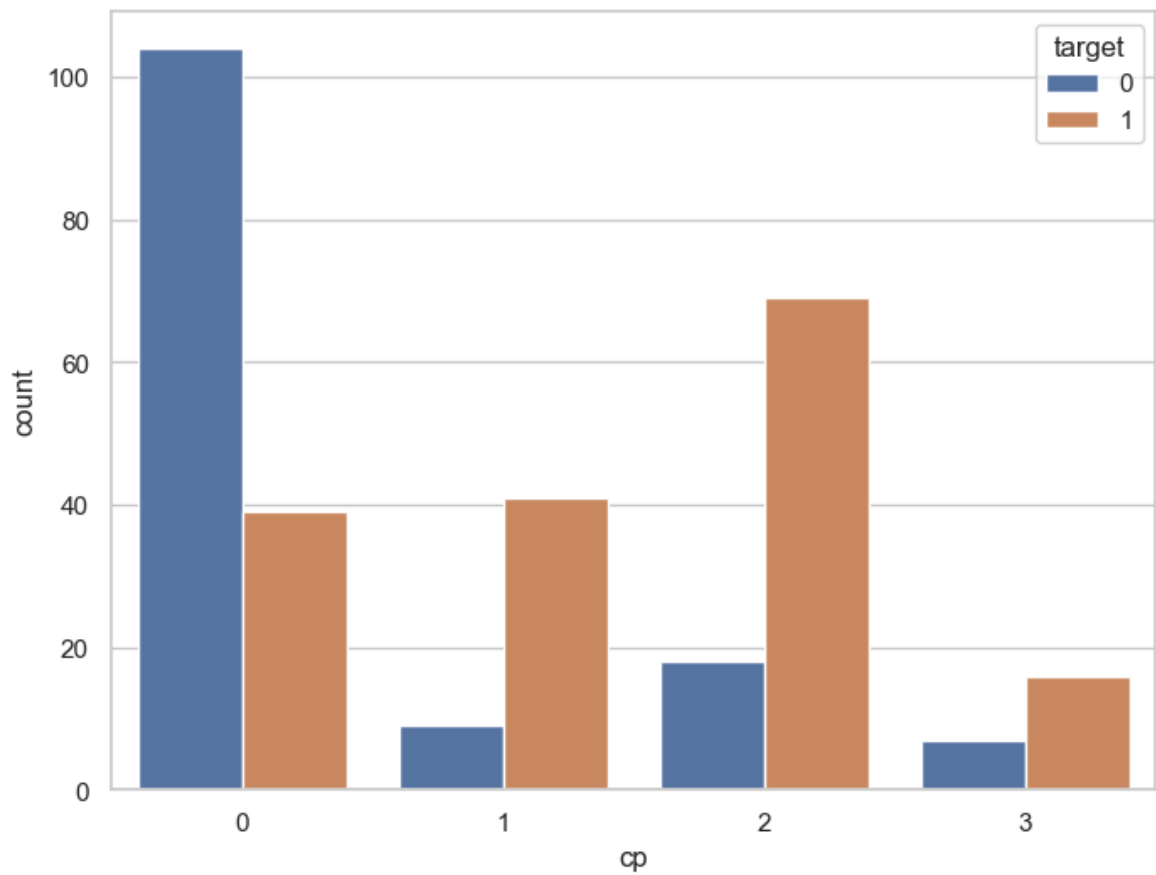
```
In [30]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.countplot(x="cp", data=df)  
plt.show()
```



```
In [31]: df.groupby('cp')['target'].value_counts()
```

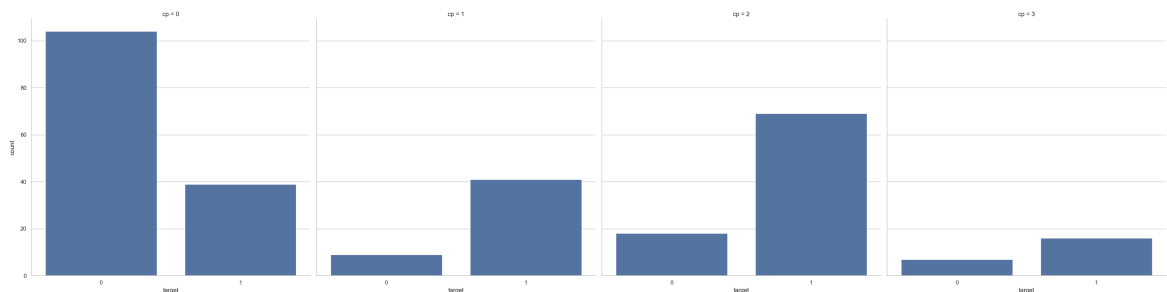
```
Out[31]: cp target
0      0      104
      1       39
1      1       41
      0        9
2      1       69
      0       18
3      1       16
      0        7
Name: count, dtype: int64
```

```
In [32]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="cp", hue="target", data=df)
plt.show()
```



```
In [33]: ax = sns.catplot(x="target", col="cp", data=df, kind="count", height=8, aspect=1
```

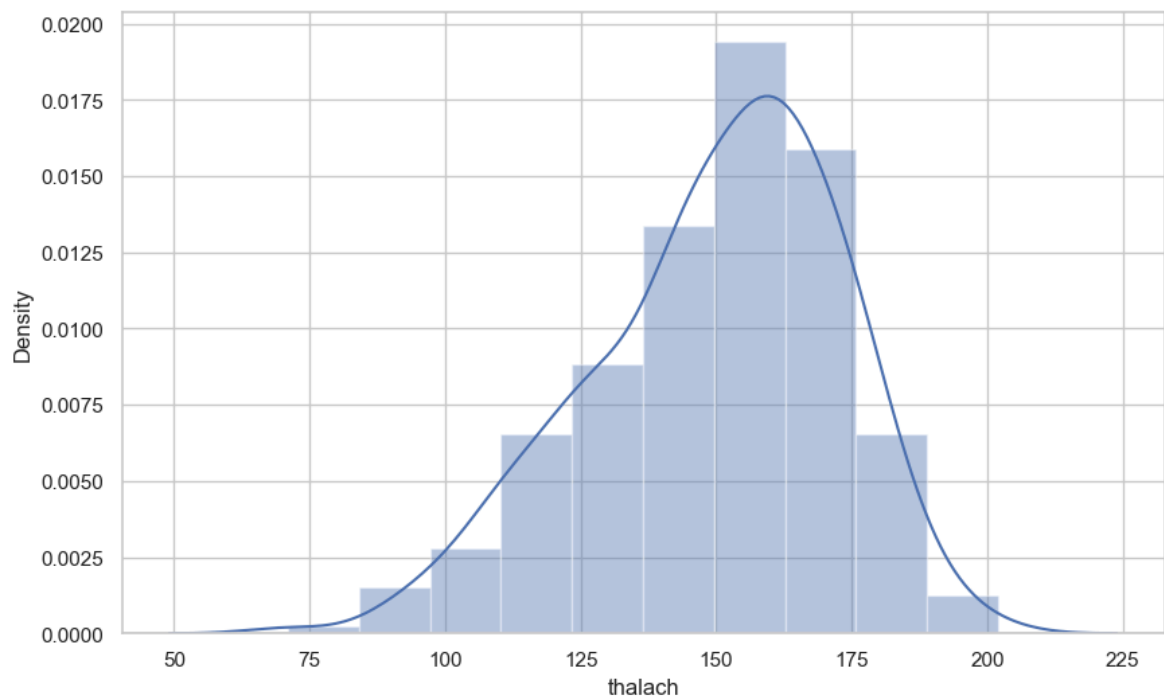
```
In [34]: plt.show()
```



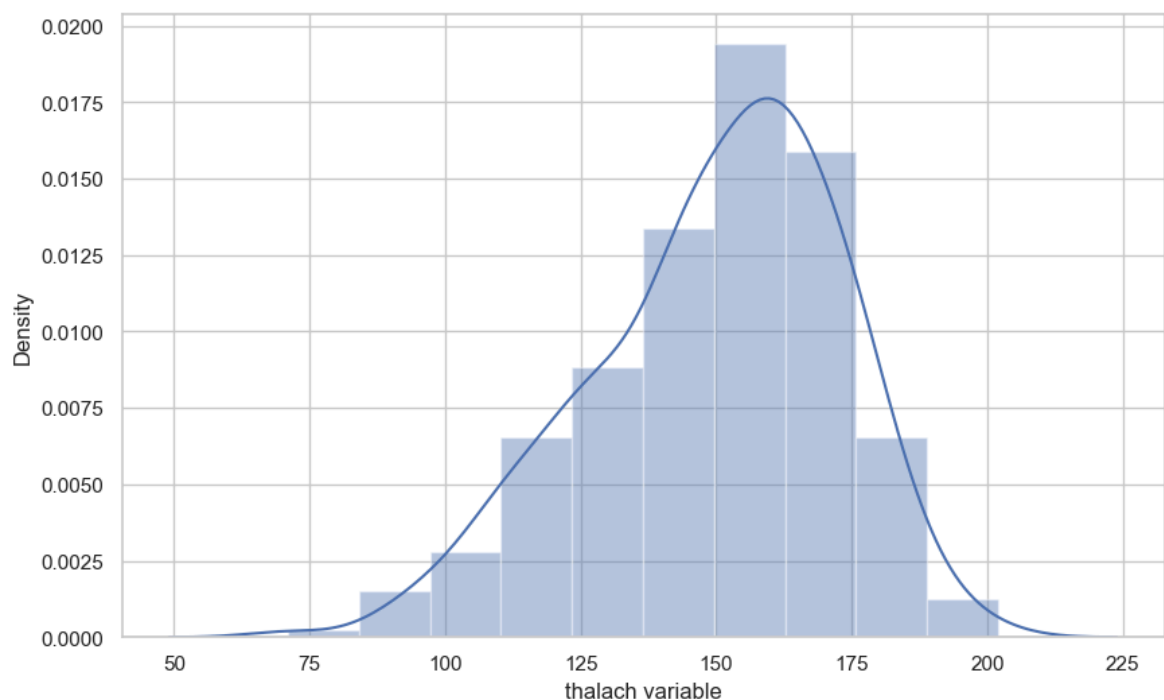
```
In [35]: df['thalach'].nunique()
```

```
Out[35]: 91
```

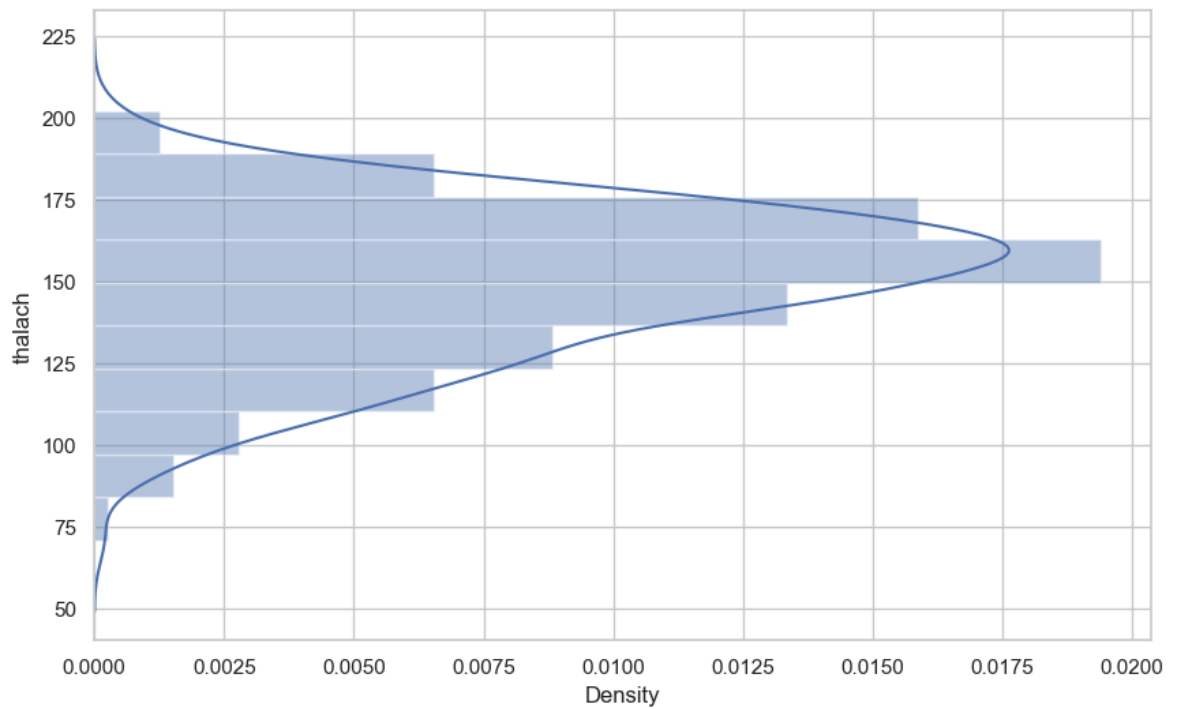
```
In [36]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
ax = sns.distplot(x, bins=10)
plt.show()
```



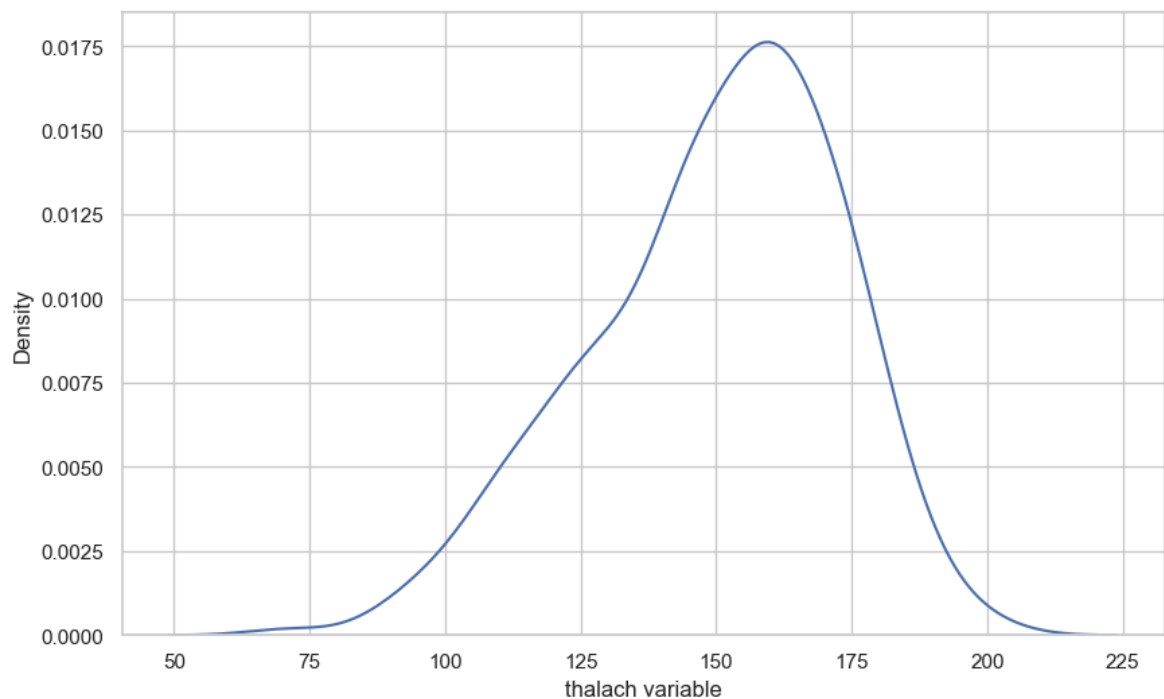
```
In [37]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
x = pd.Series(x, name="thalach variable")
ax = sns.distplot(x, bins=10)
plt.show()
```



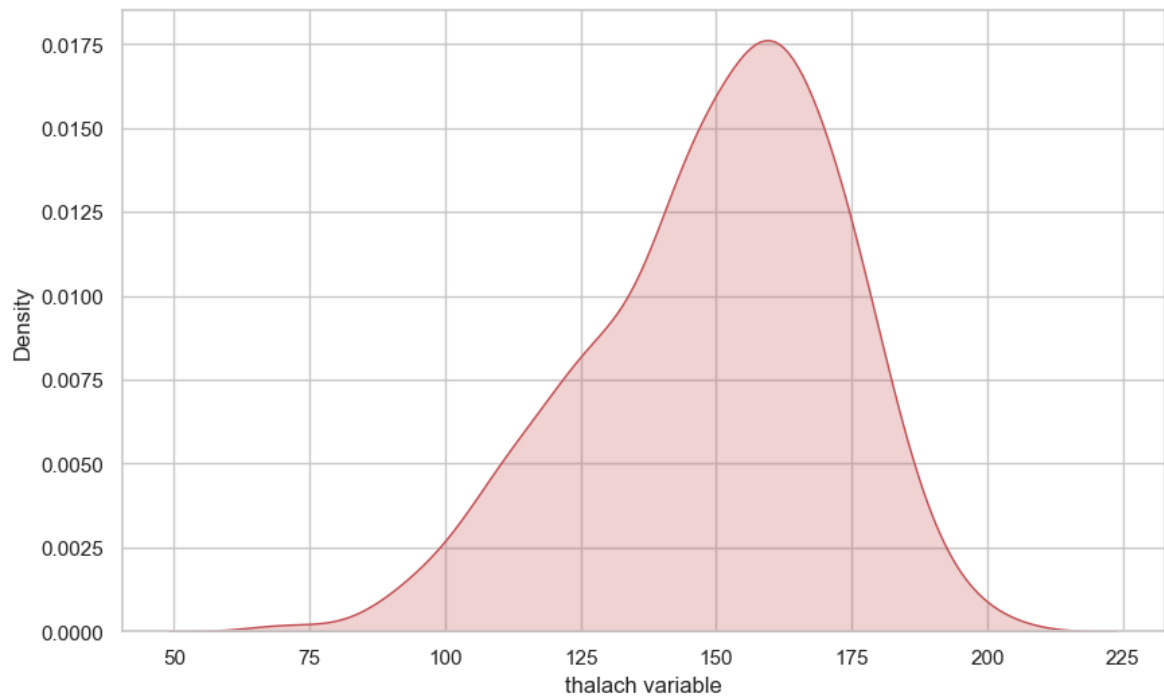
```
In [38]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
ax = sns.distplot(x, bins=10, vertical=True)
plt.show()
```



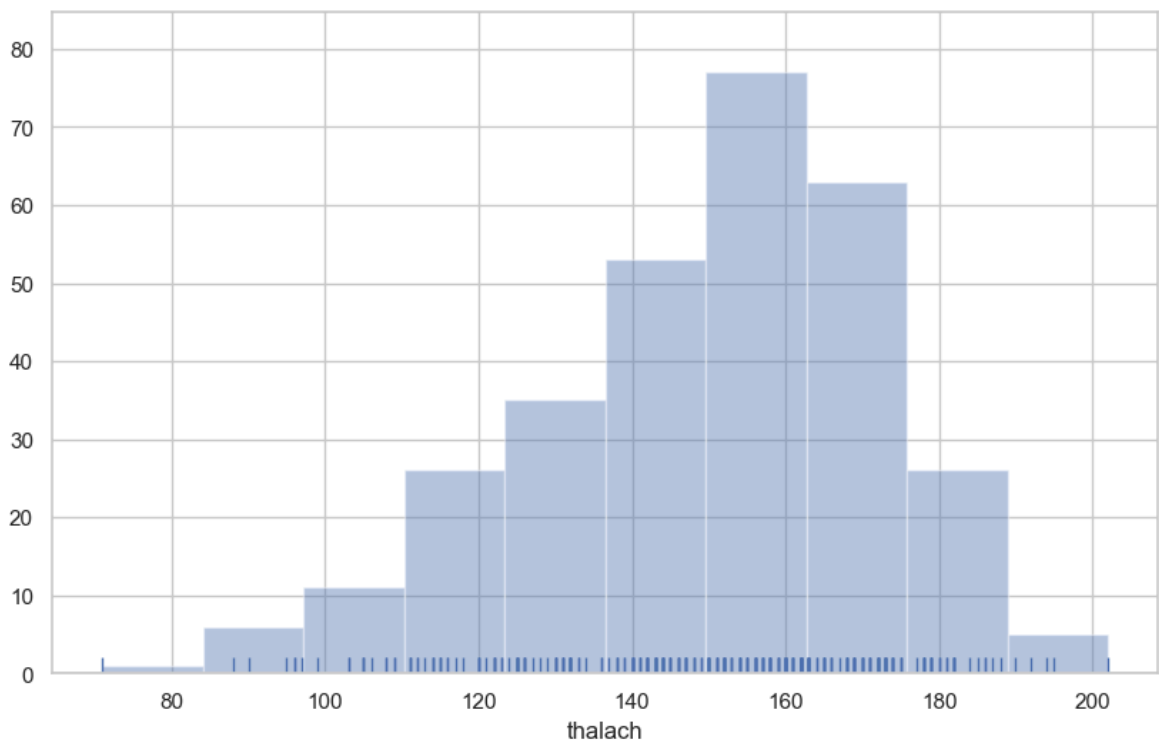
```
In [39]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
x = pd.Series(x, name="thalach variable")
ax = sns.kdeplot(x)
plt.show()
```



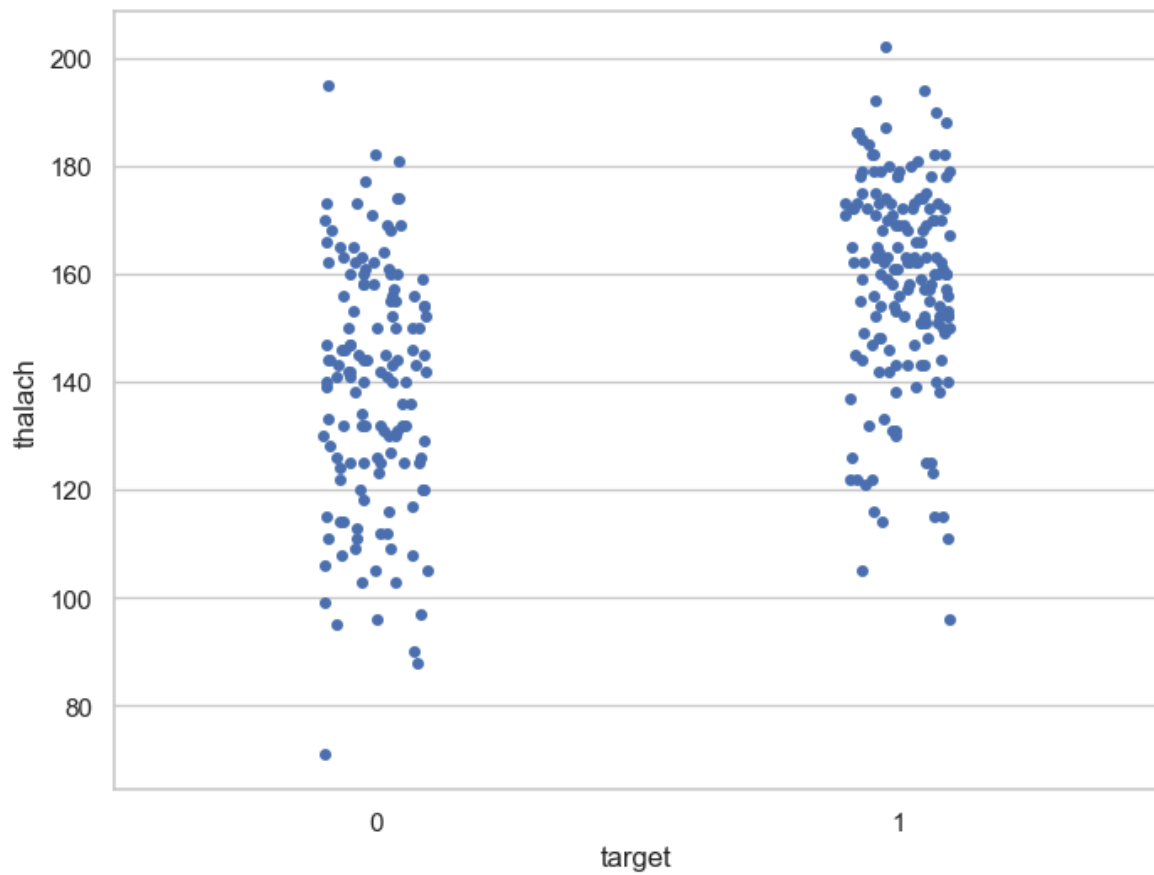
```
In [40]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
x = pd.Series(x, name="thalach variable")
ax = sns.kdeplot(x, shade=True, color='r')
plt.show()
```



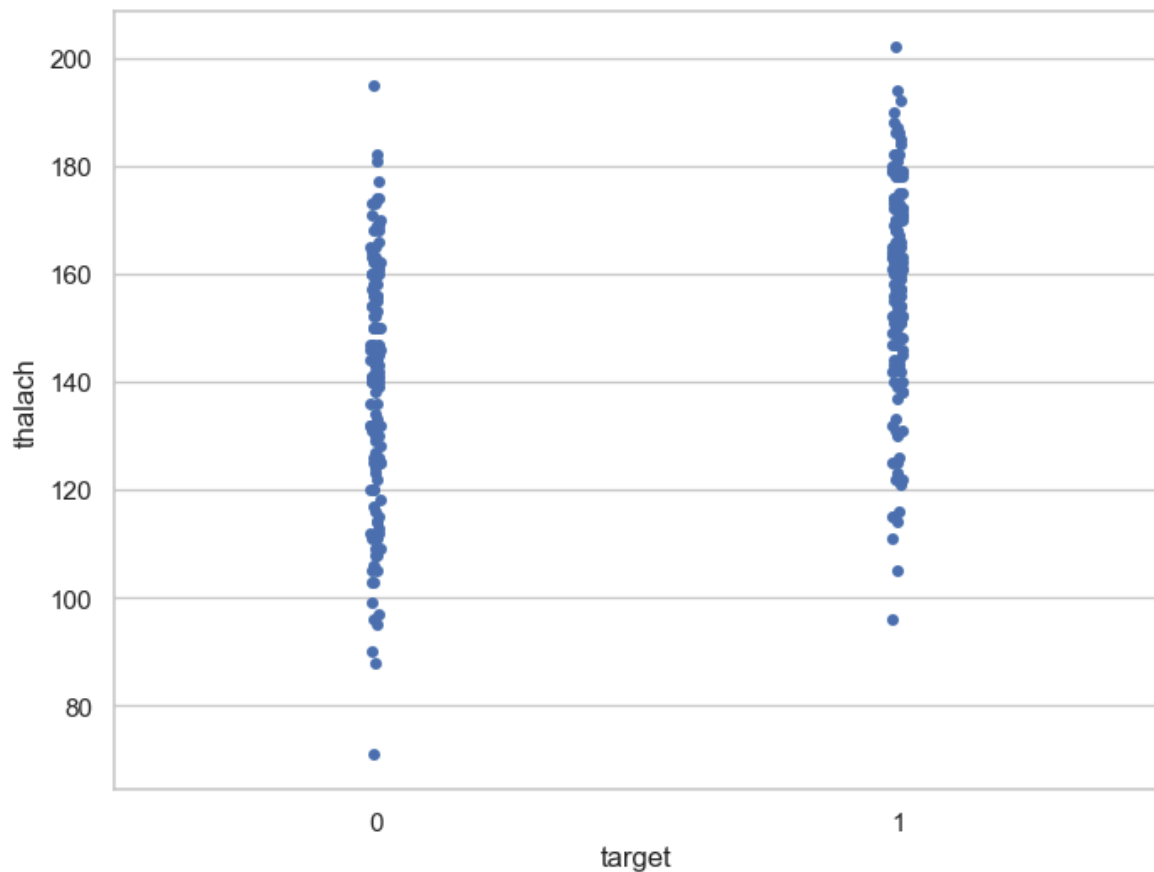
```
In [41]: f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
ax = sns.distplot(x, kde=False, rug=True, bins=10)
plt.show()
```



```
In [42]: f, ax = plt.subplots(figsize=(8, 6))
sns.stripplot(x="target", y="thalach", data=df)
plt.show()
```

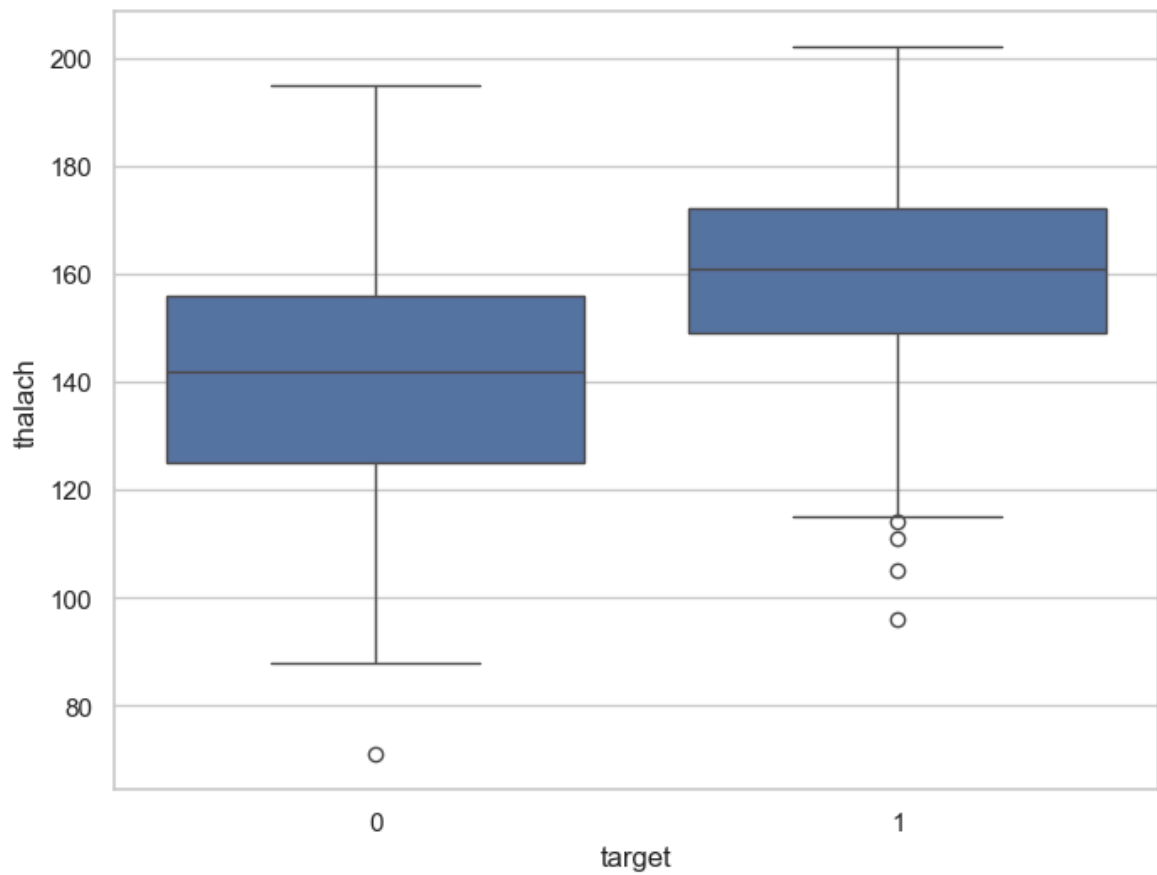


```
In [43]: f, ax = plt.subplots(figsize=(8, 6))  
sns.stripplot(x="target", y="thalach", data=df, jitter = 0.01)  
plt.show()
```

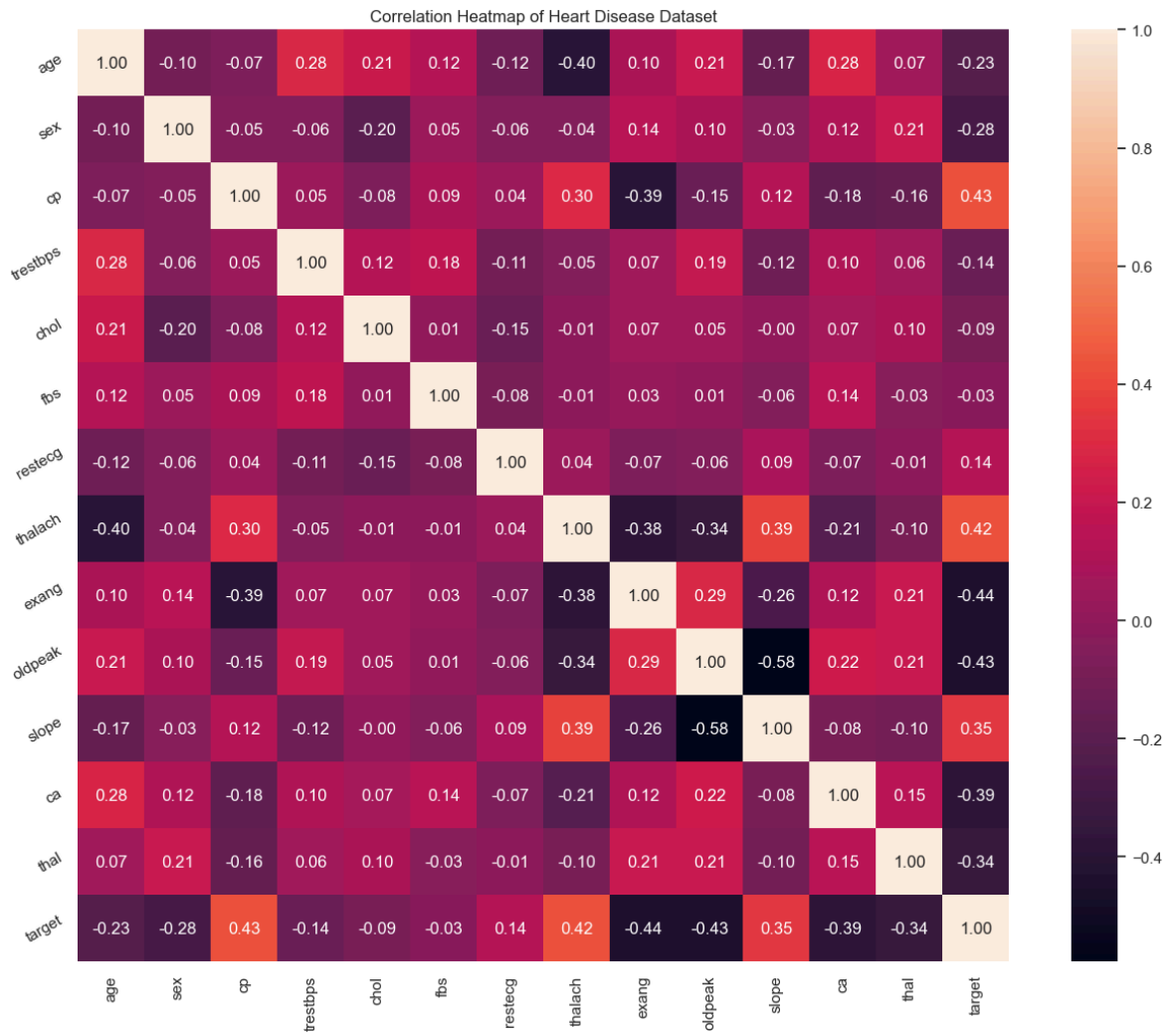


```
In [44]: f, ax = plt.subplots(figsize=(8, 6))  
sns.boxplot(x="target", y="thalach", data=df)
```

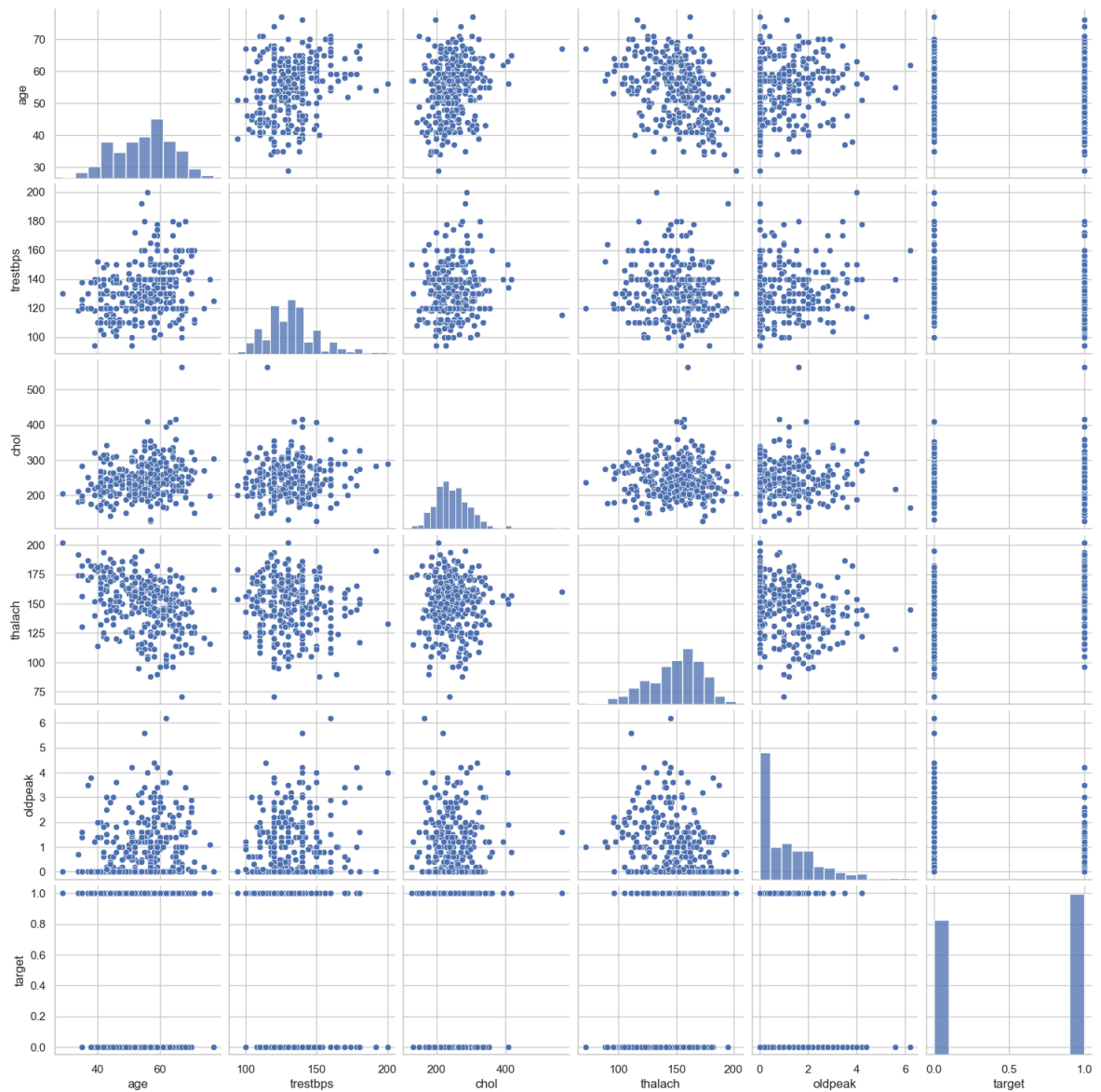
```
plt.show()
```



```
In [45]: plt.figure(figsize=(16,12))
plt.title('Correlation Heatmap of Heart Disease Dataset')
a = sns.heatmap(correlation, square=True, annot=True, fmt='.2f', linecolor='white')
a.set_xticklabels(a.get_xticklabels(), rotation=90)
a.set_yticklabels(a.get_yticklabels(), rotation=30)
plt.show()
```

```
In [46]: num_var = ['age', 'trestbps', 'chol', 'thalach', 'oldpeak', 'target' ]
sns.pairplot(df[num_var], kind='scatter', diag_kind='hist')
plt.show()
```



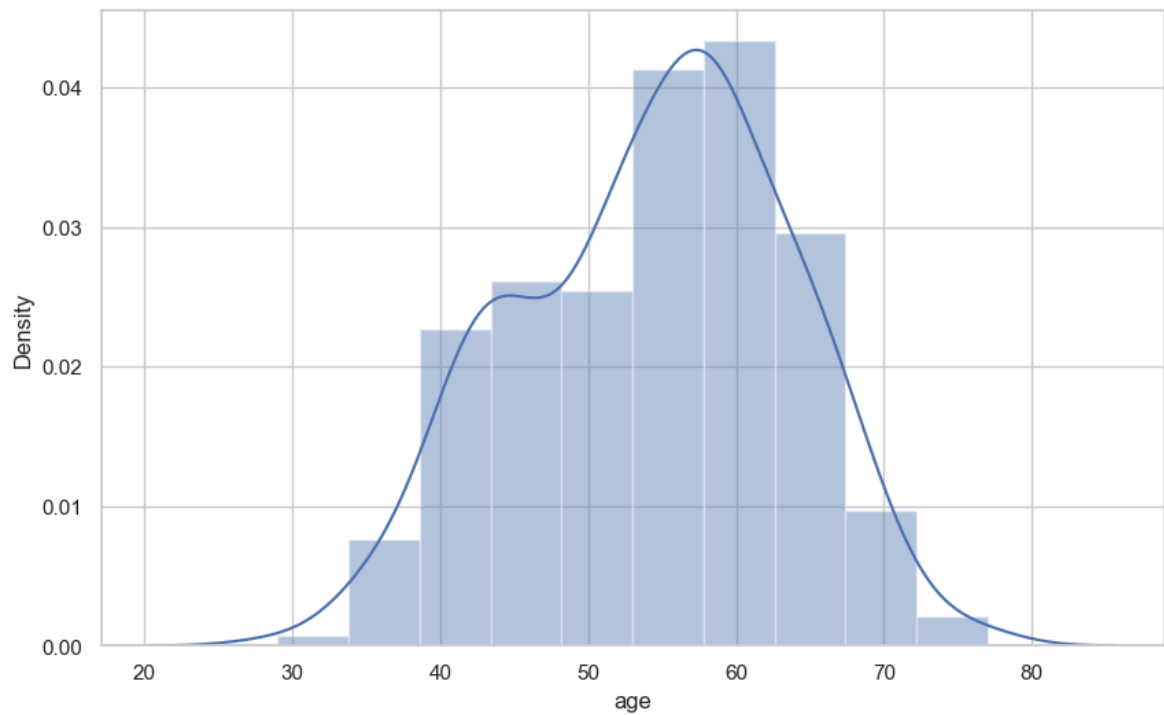
```
In [47]: df['age'].unique()
```

```
Out[47]: 41
```

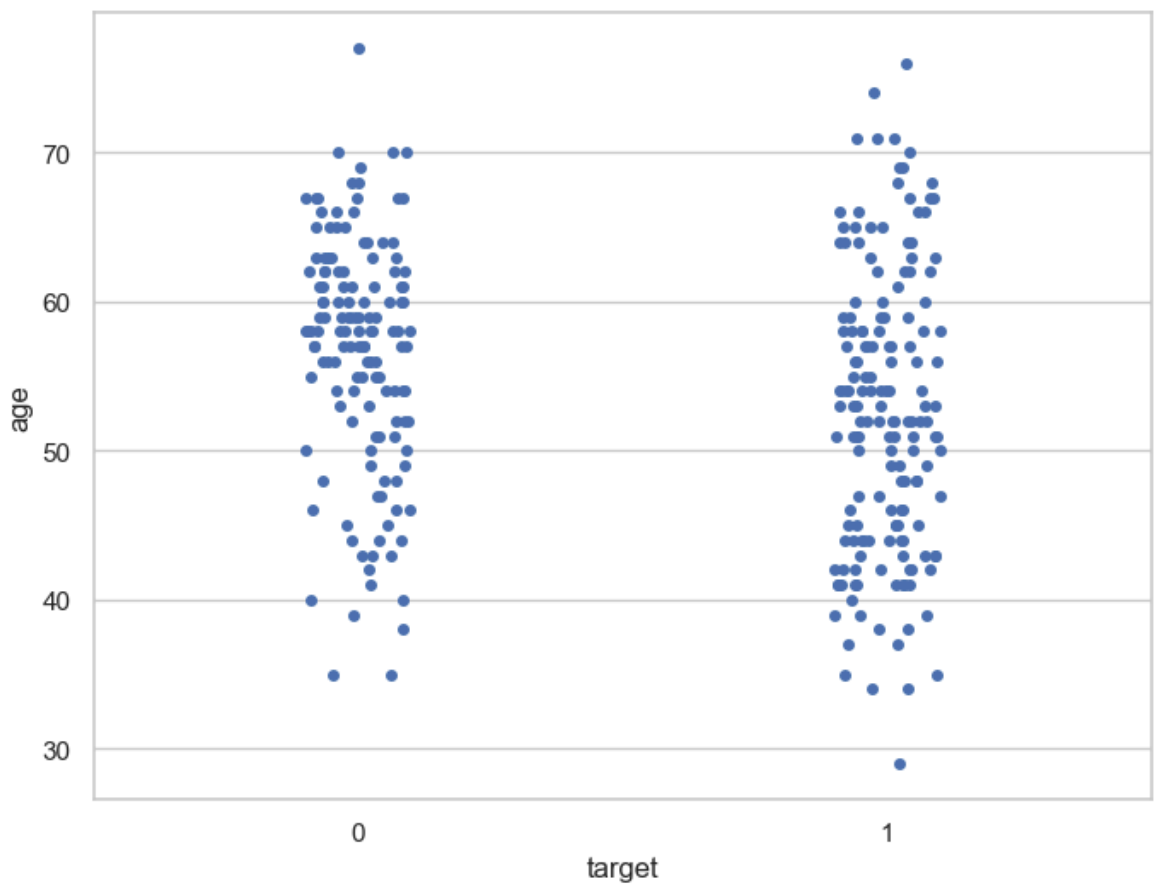
```
In [48]: df['age'].describe()
```

```
Out[48]: count    303.000000
mean       54.366337
std        9.082101
min        29.000000
25%        47.500000
50%        55.000000
75%        61.000000
max        77.000000
Name: age, dtype: float64
```

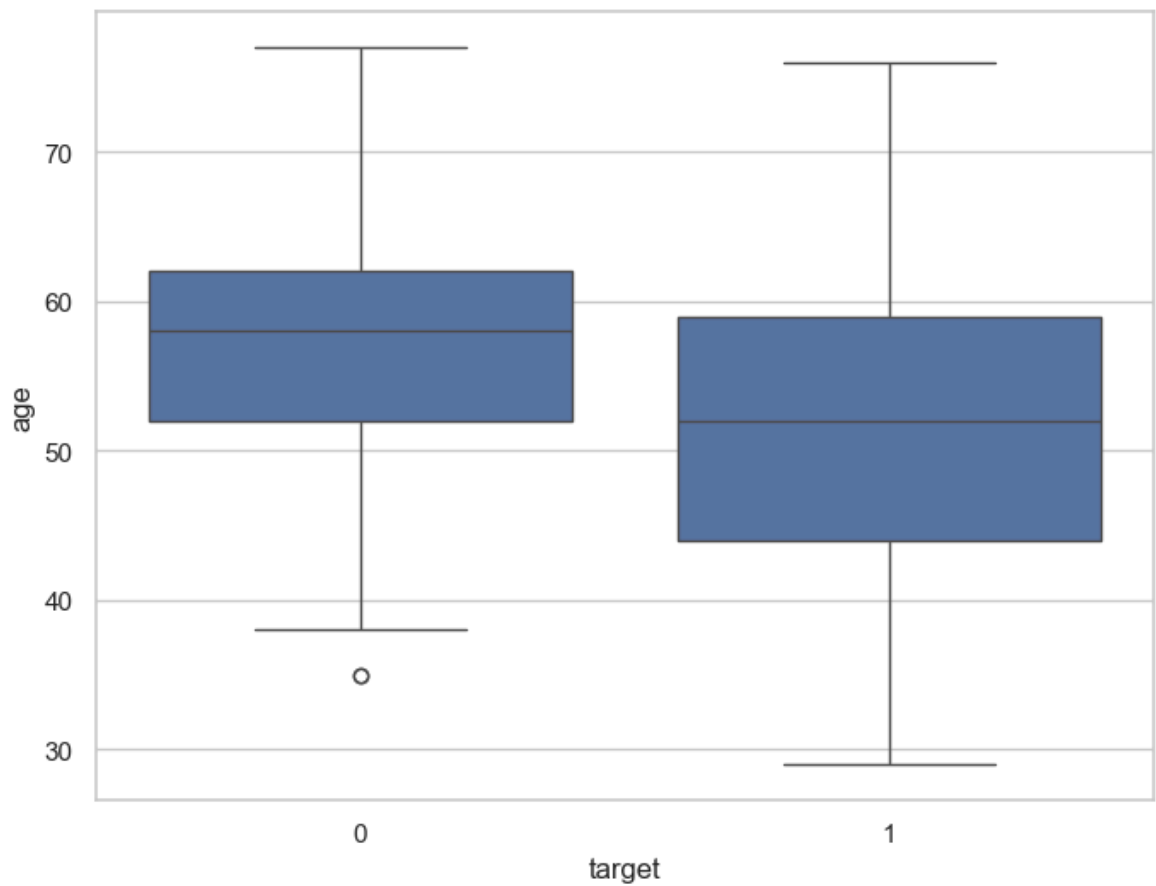
```
In [49]: f, ax = plt.subplots(figsize=(10,6))
x = df['age']
ax = sns.distplot(x, bins=10)
plt.show()
```



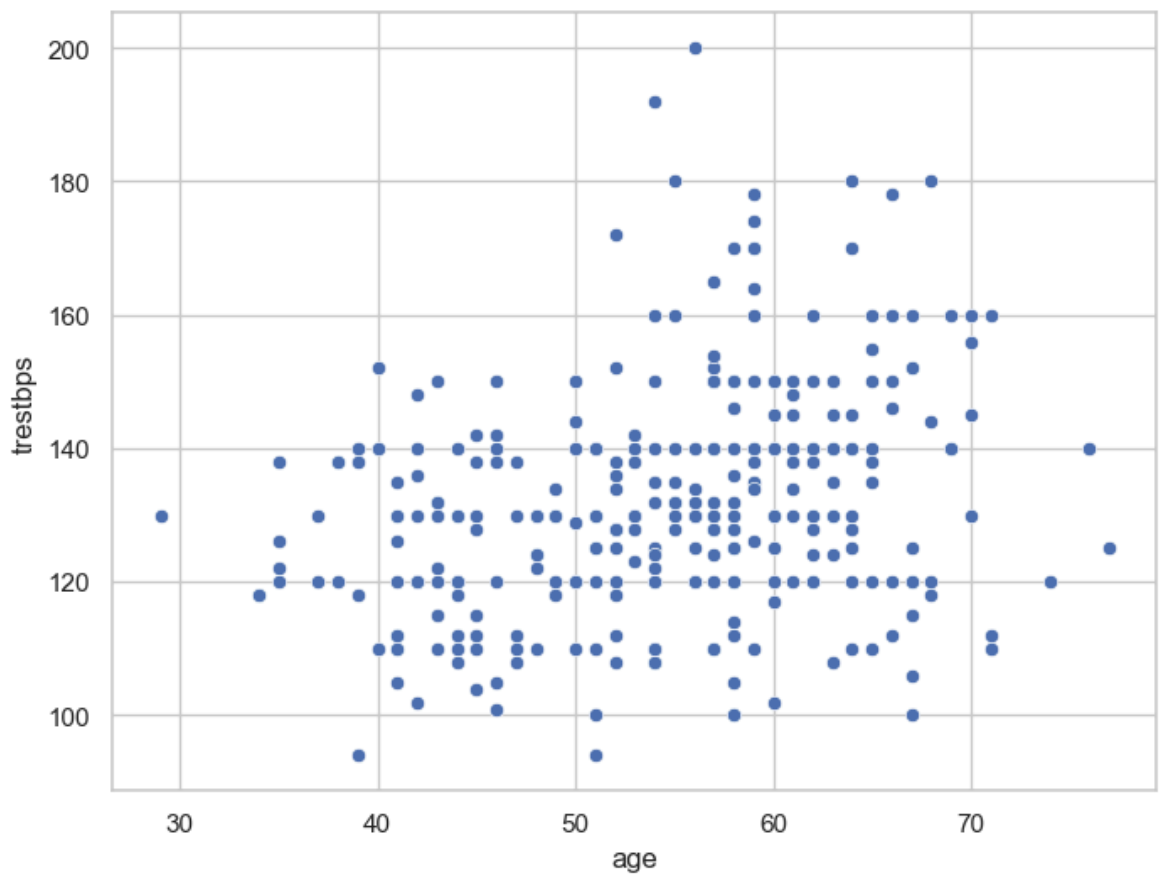
```
In [50]: f, ax = plt.subplots(figsize=(8, 6))
sns.stripplot(x="target", y="age", data=df)
plt.show()
```



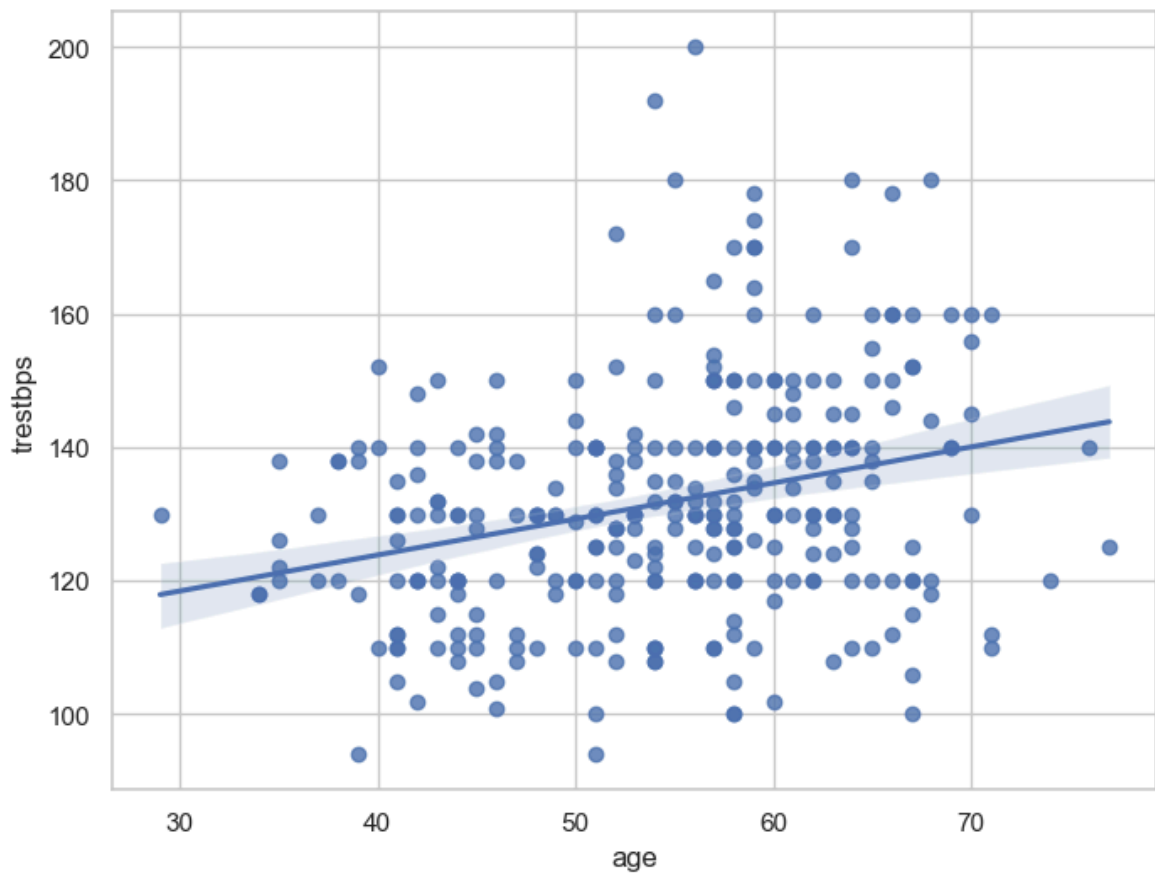
```
In [51]: f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x="target", y="age", data=df)
plt.show()
```



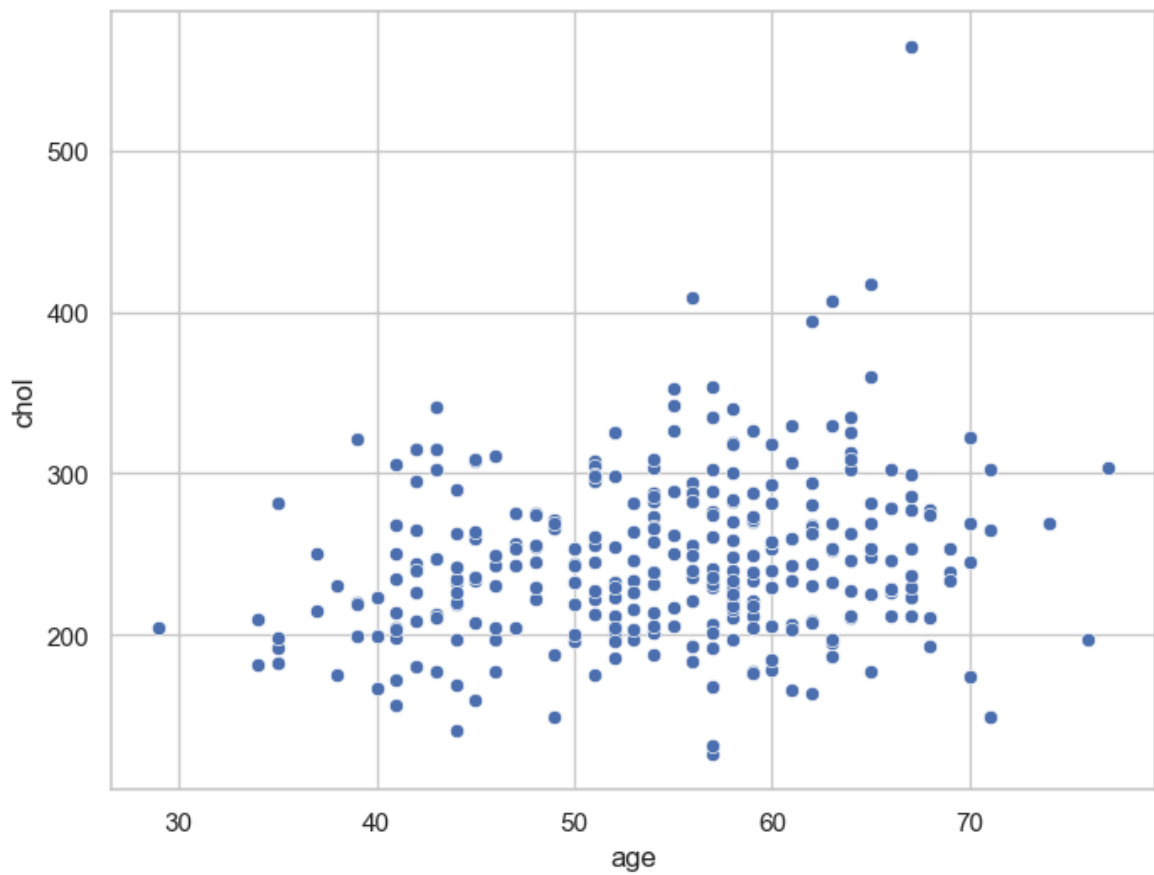
```
In [52]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.scatterplot(x="age", y="trestbps", data=df)
plt.show()
```



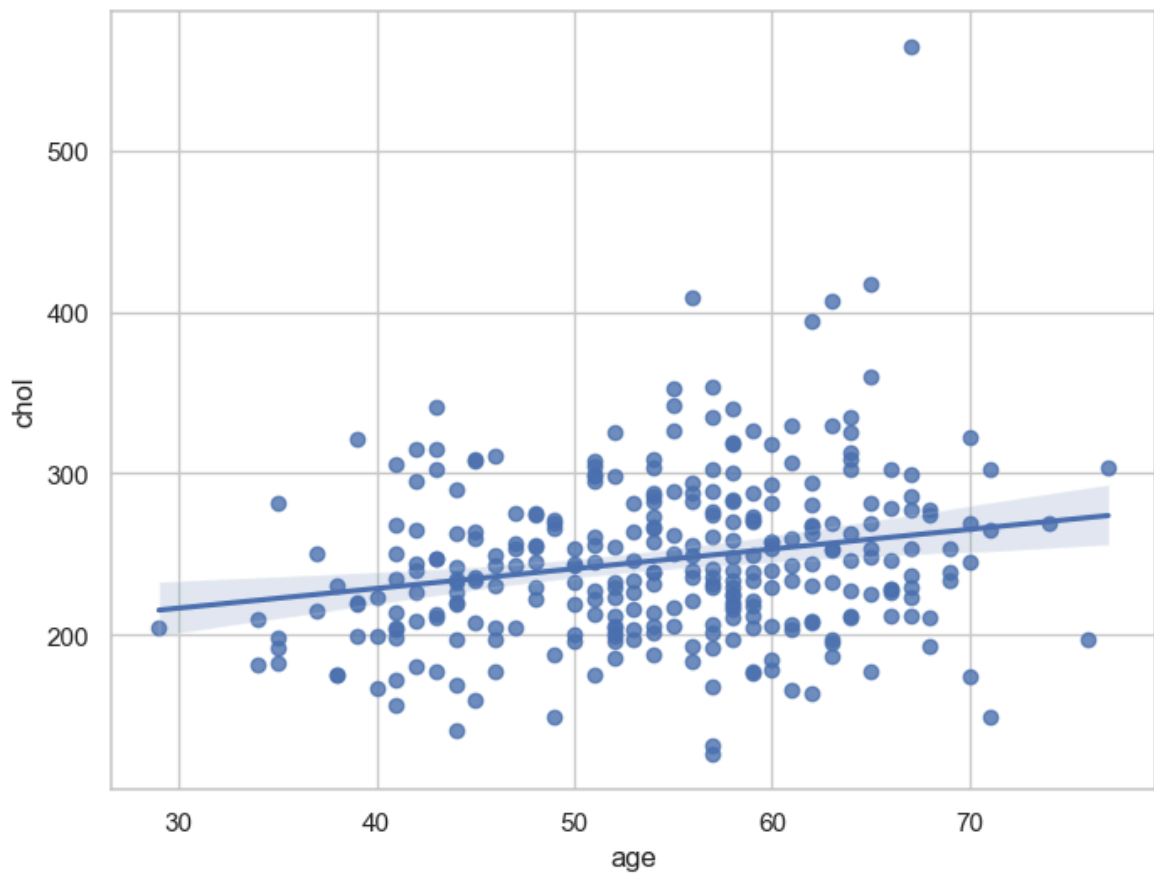
```
In [53]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.regplot(x="age", y="trestbps", data=df)
plt.show()
```



```
In [54]: f, ax = plt.subplots(figsize=(8, 6))
ax = sns.scatterplot(x="age", y="chol", data=df)
plt.show()
```

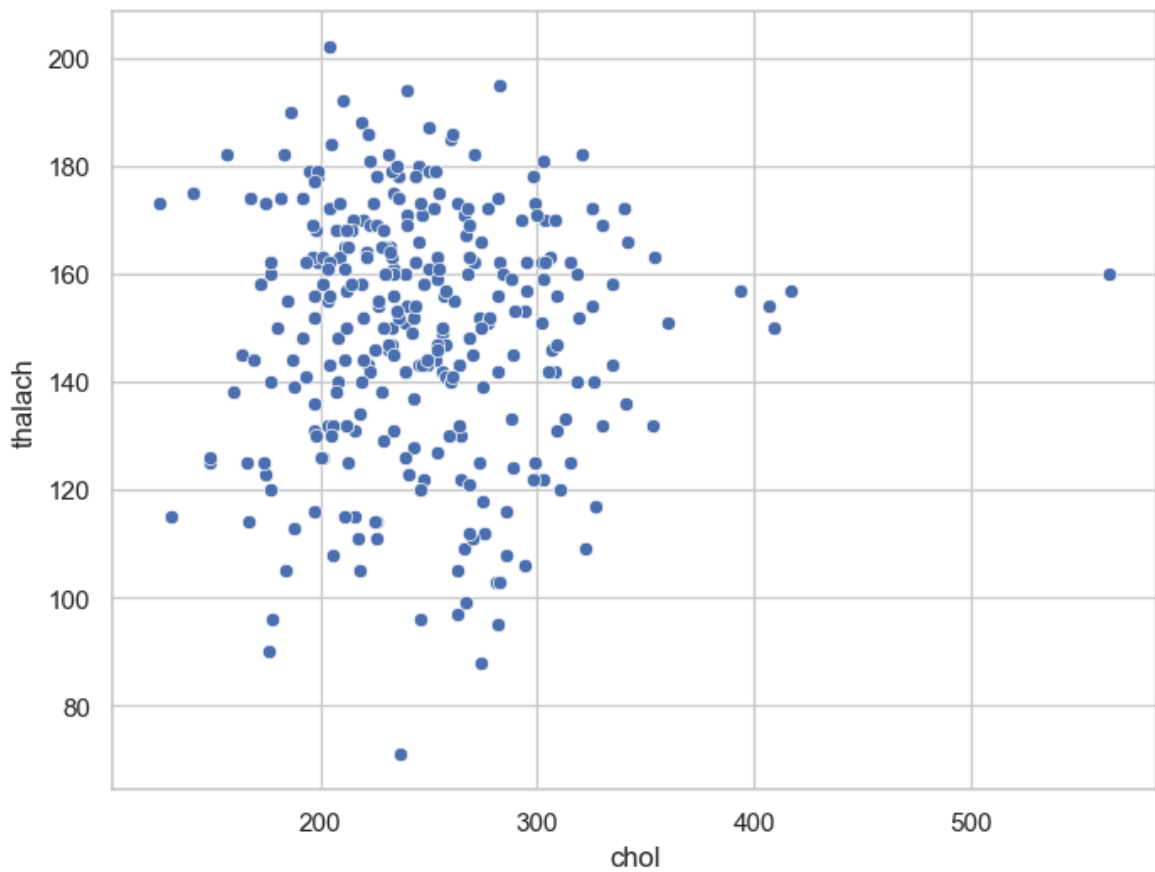


```
In [55]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.regplot(x="age", y="chol", data=df)  
plt.show()
```

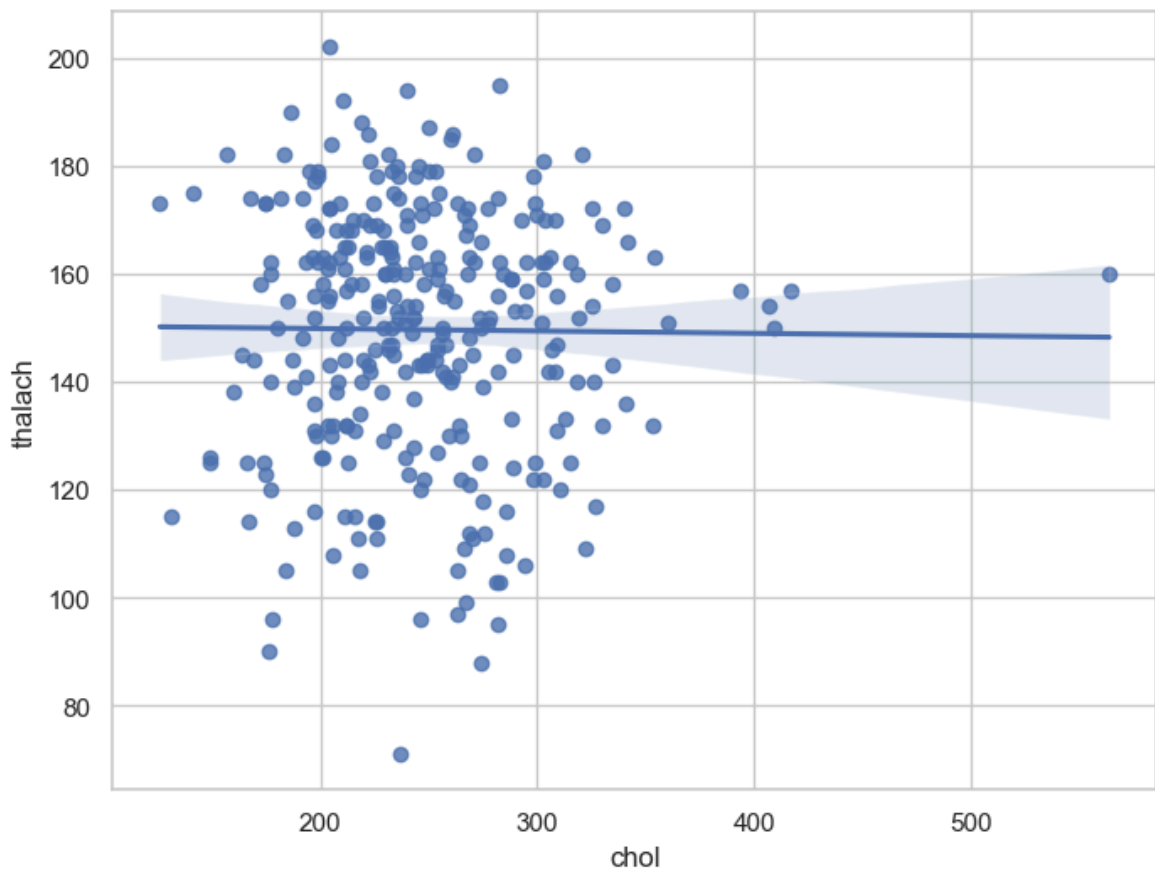


```
In [56]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.scatterplot(x="chol", y = "thalach", data=df)
```

```
plt.show()
```



```
In [57]: f, ax = plt.subplots(figsize=(8, 6))  
ax = sns.regplot(x="chol", y="thalach", data=df)  
plt.show()
```



```
In [58]: df.isnull().sum()
```

```
Out[58]: age          0
sex          0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          0
thal        0
target      0
dtype: int64
```

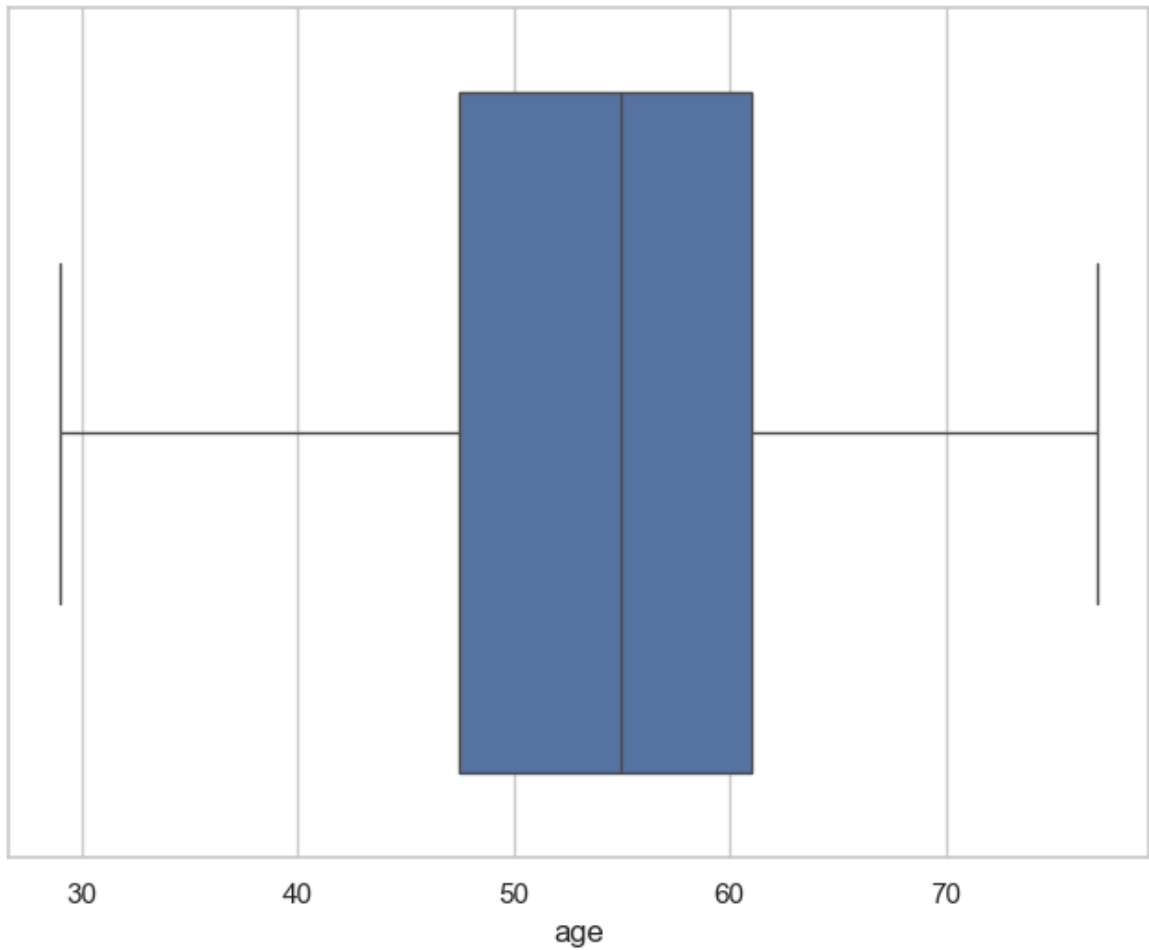
```
In [59]: assert pd.notnull(df).all().all()
```

```
In [60]: assert (df >= 0).all().all()
```

```
In [61]: df['age'].describe()
```

```
Out[61]: count    303.000000
mean      54.366337
std       9.082101
min       29.000000
25%      47.500000
50%      55.000000
75%      61.000000
max       77.000000
Name: age, dtype: float64
```

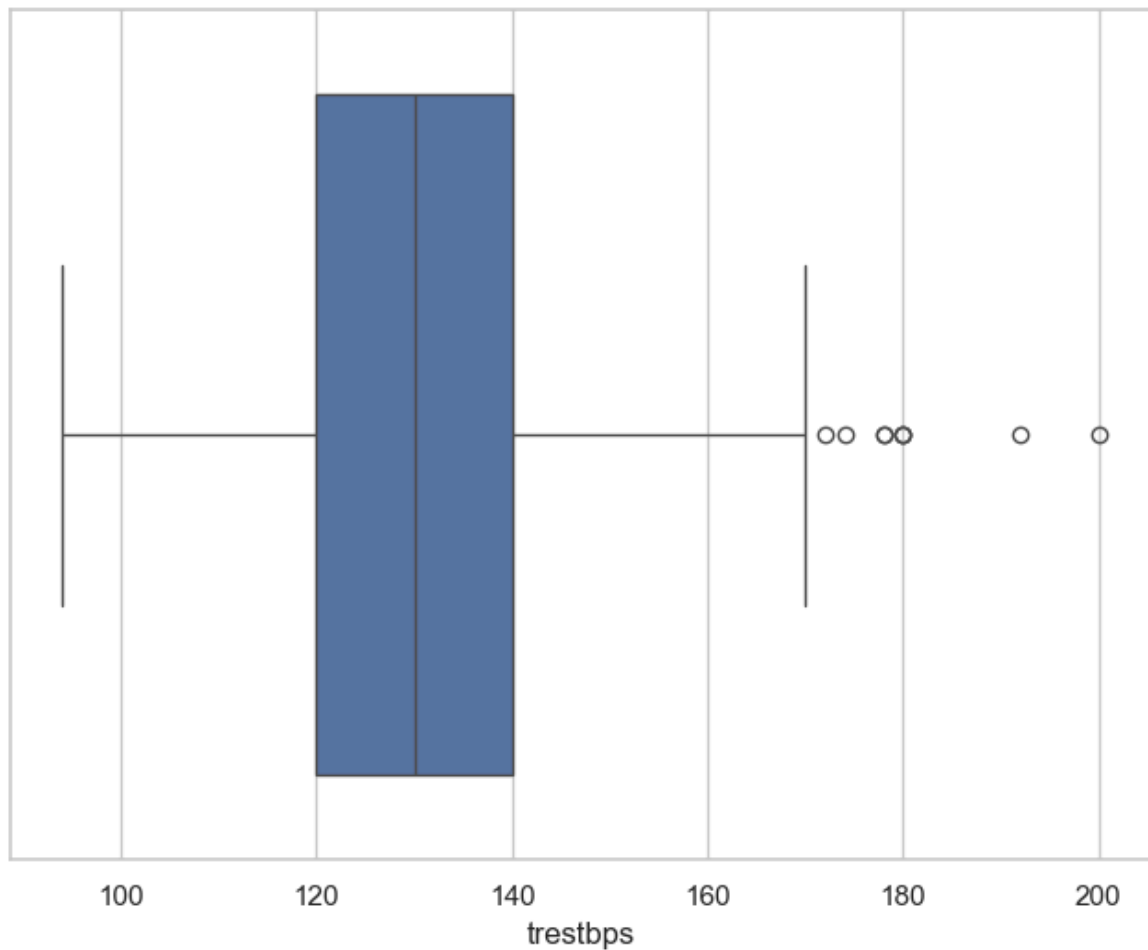
```
In [62]: f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x=df["age"])
plt.show()
```

```
In [63]: df['trestbps'].describe()
```

```
Out[63]: count    303.000000  
mean      131.623762  
std       17.538143  
min       94.000000  
25%      120.000000  
50%      130.000000  
75%      140.000000  
max      200.000000  
Name: trestbps, dtype: float64
```

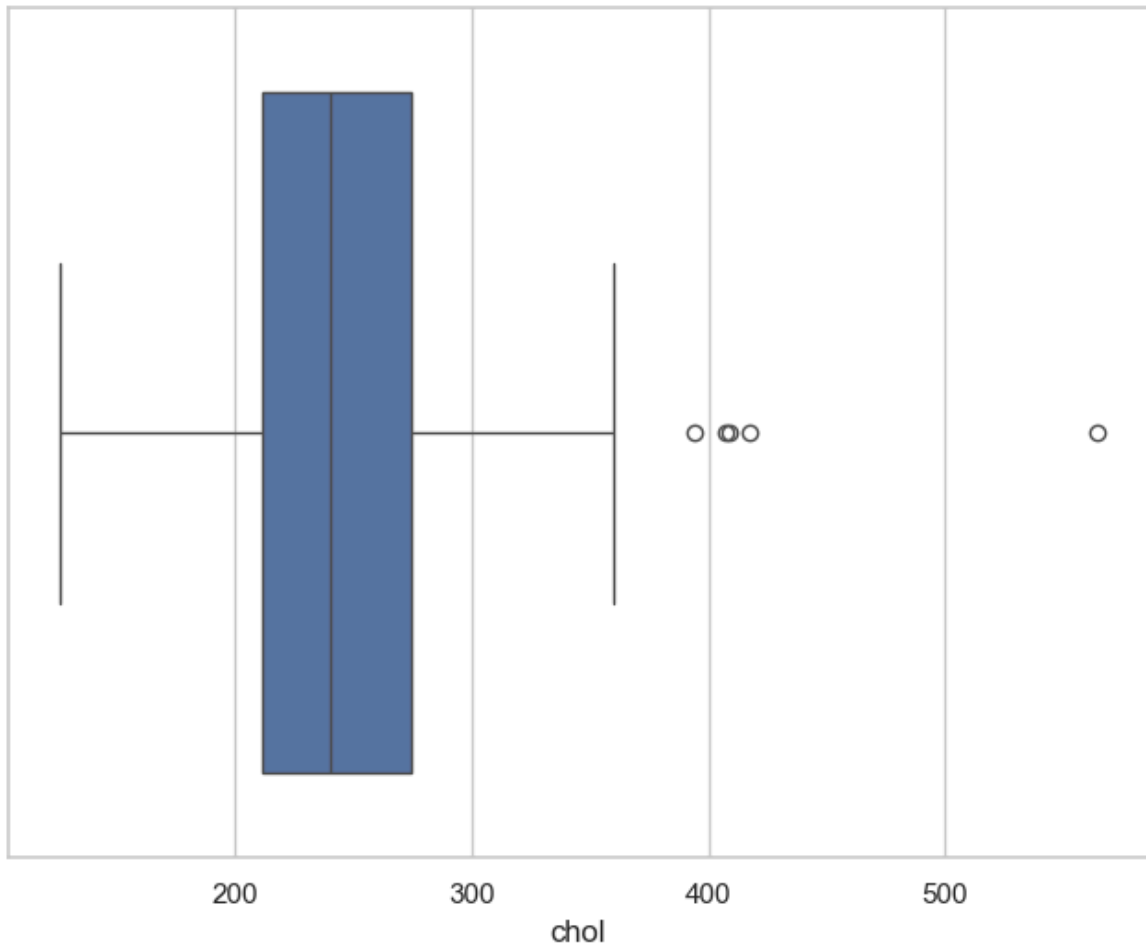
```
In [64]: f, ax = plt.subplots(figsize=(8, 6))  
sns.boxplot(x=df["trestbps"])  
plt.show()
```



```
In [65]: df['chol'].describe()
```

```
Out[65]: count    303.000000  
mean      246.264026  
std       51.830751  
min       126.000000  
25%      211.000000  
50%      240.000000  
75%      274.500000  
max       564.000000  
Name: chol, dtype: float64
```

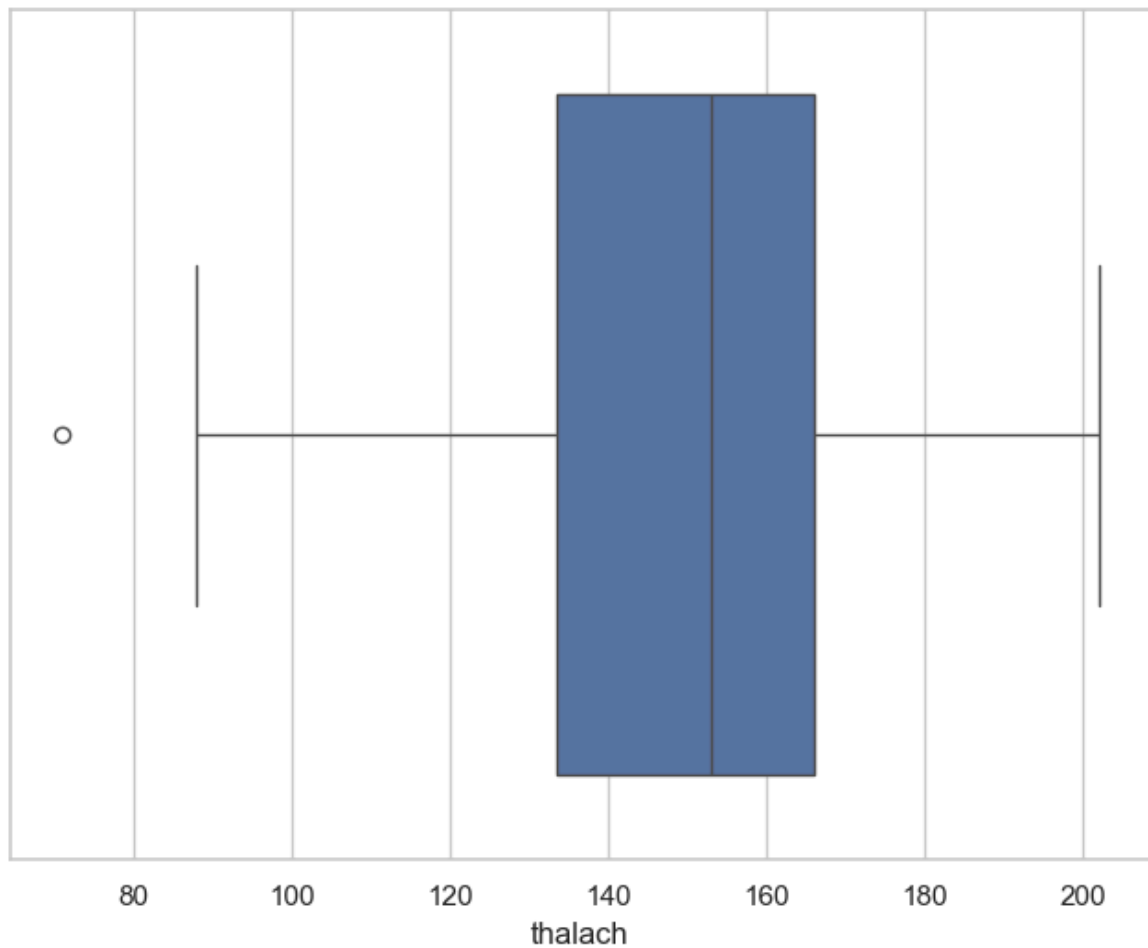
```
In [66]: f, ax = plt.subplots(figsize=(8, 6))  
sns.boxplot(x=df["chol"])  
plt.show()
```



```
In [67]: df['thalach'].describe()
```

```
Out[67]: count    303.000000
mean      149.646865
std       22.905161
min       71.000000
25%      133.500000
50%      153.000000
75%      166.000000
max       202.000000
Name: thalach, dtype: float64
```

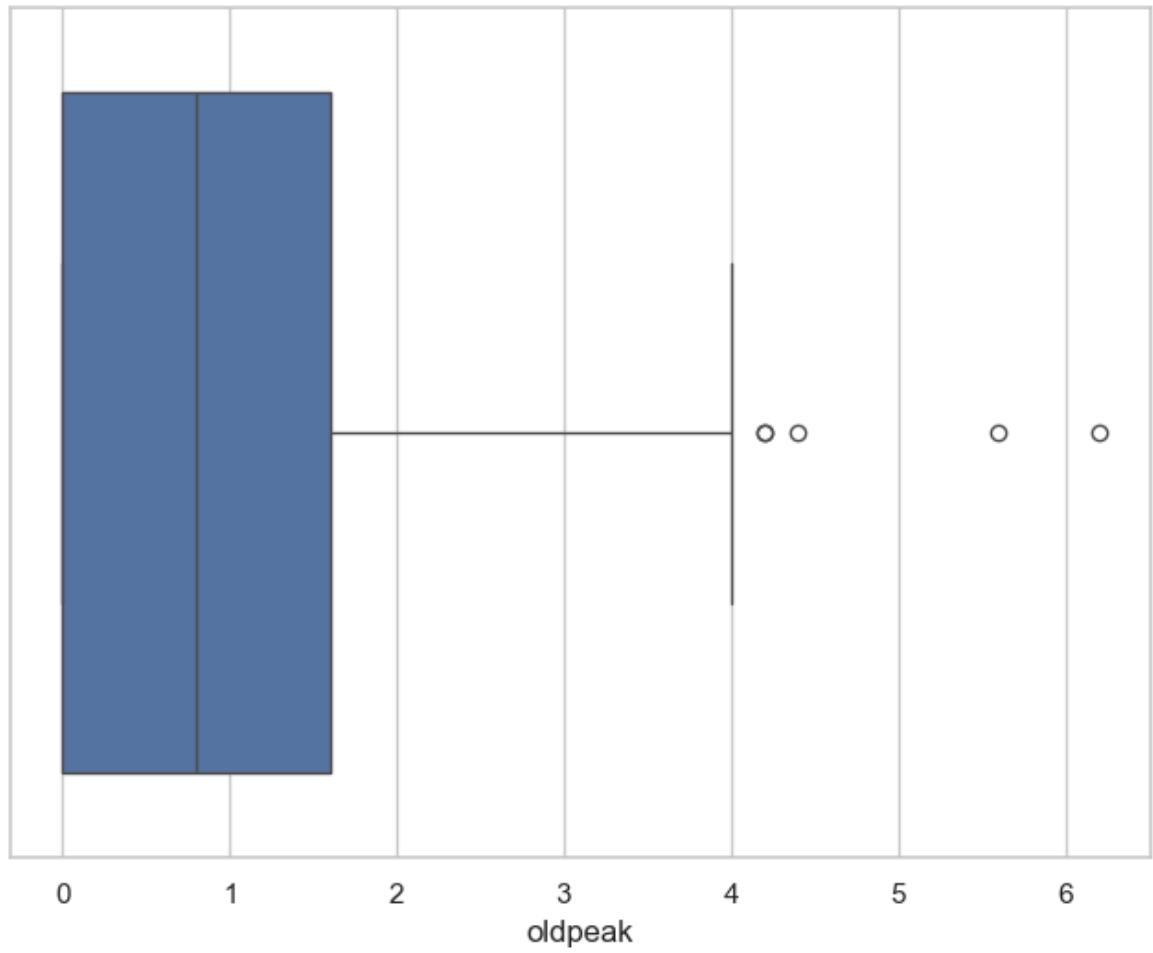
```
In [68]: f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x=df["thalach"])
plt.show()
```



```
In [69]: df['oldpeak'].describe()
```

```
Out[69]: count    303.000000  
mean         1.039604  
std          1.161075  
min          0.000000  
25%          0.000000  
50%          0.800000  
75%          1.600000  
max          6.200000  
Name: oldpeak, dtype: float64
```

```
In [70]: f, ax = plt.subplots(figsize=(8, 6))  
sns.boxplot(x=df["oldpeak"])  
plt.show()
```



In []: